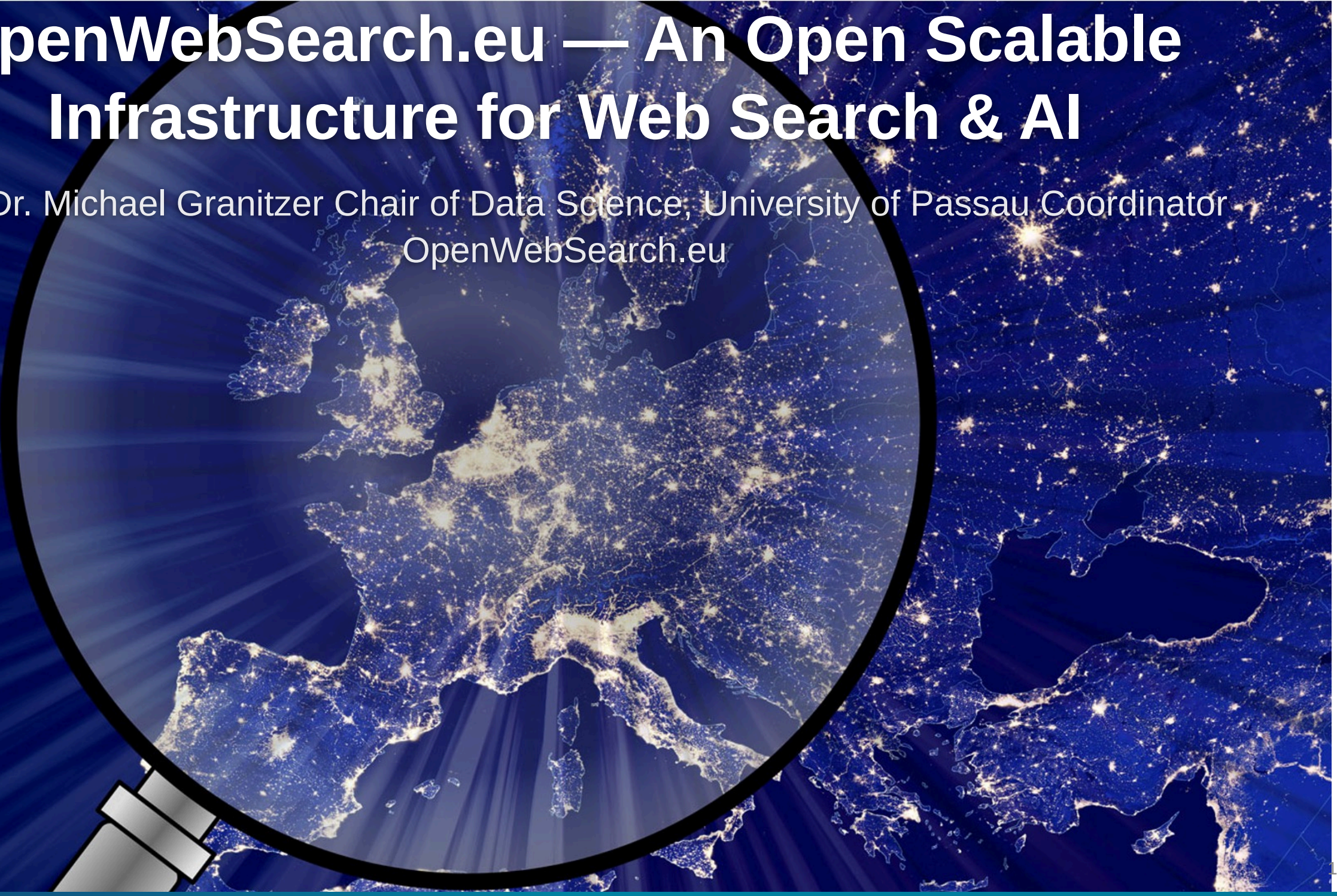


# OpenWebSearch.eu — An Open Scalable Infrastructure for Web Search & AI

Prof. Dr. Michael Granitzer Chair of Data Science, University of Passau Coordinator  
OpenWebSearch.eu



# Web Search Today

A critical resource managed as an oligopoly

## Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



# But does it matter? Yes — on four dimensions

## User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

## Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

## Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

## AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

Being able to navigate the Web is as fundamental as GPS or the power grid

# Challenges

## THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

### VOLUME

1.15 Billion

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

Top 0.1%–1% of sites

may account for roughly half of all pages  
heavy-tail intuition from large web graph studies

~ 400 Billion

documents in Google's search index  
2020 figure cited in US v. Google testimony

Sites ≠ domains ≠ pages; use exact counts cautiously

### VELOCITY

+8.6 M / month

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

~ 3.2 / second

same monthly net change, averaged across December

149 ZB (global)

all digital data created / copied worldwide in 2024  
IDC / Statista global datasphere estimate

The web changes constantly; crawling is continuous, not one-off

### VARIETY

150+ languages

observed across websites

W3Techs usage survey

249 CMS Platforms

CMS platforms identified in the HTTP Archive dataset  
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

HTML · PDF · JSON-LD · media · APIs

content types to parse and process

HTML · PDFs · JS apps · schemas · media · APIs

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·  
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista



- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

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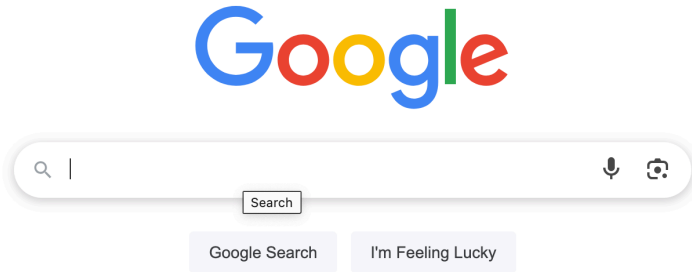
High upfront costs to tap the Web as resources  
— especially small innovators and researchers  
are left behind

# WHY IT MATTERS - IN MORE DETAILS

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## Reduced user choice

Single source of our information: We would not be happy with only one European newspaper



Newspaper map: [reddit.com/r/MapPorn/comments/1c70v2q/newspapers\\_of\\_record\\_or\\_note\\_in\\_europe/](https://reddit.com/r/MapPorn/comments/1c70v2q/newspapers_of_record_or_note_in_europe/)

## Search Engine Manipulation Effect

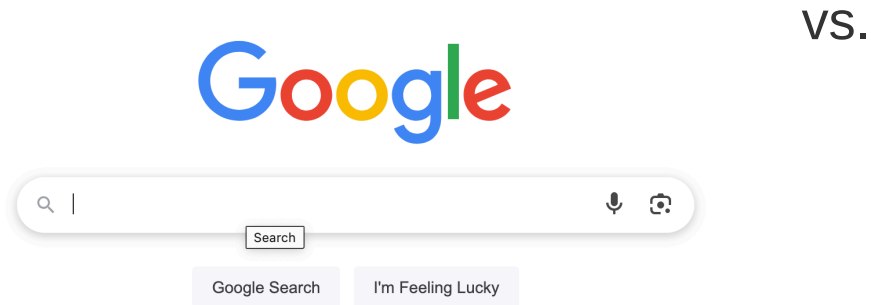
- Users locked into a single information gatekeeper
- **Search Engine Manipulation Effect (SEME)**: biased search rankings can shift voting preferences of undecided voters by **20% or more** — up to **72%** in some demographics — while users remain largely unaware of the manipulation





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# Search Engine Manipulation Effect

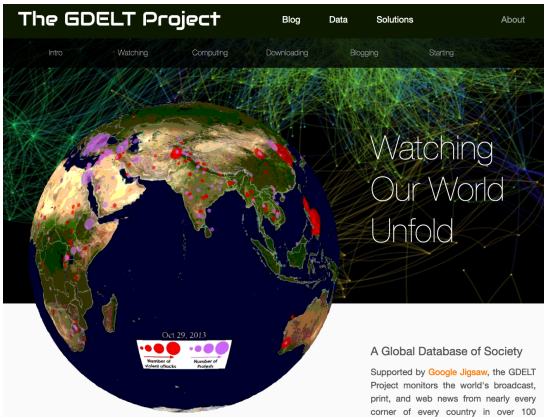
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# The web is the richest information source

The Web is the richest information source, but we are blind on what is there

- Companies, products, scientific information, events, geo-location data
- 60% of web pages contain structured data in form of Microdata / Microformats
  - Products, FAQ, Places, Shops
  - Job postings, Reviews, Events



Samsung GARMIN Fossil Apple Fitbit Price Reviews

Sponsored :

|                                                                                                                                                                                                                           |                                                                                                                                                                                                                         |                                                                                                                                                                                                                                             |                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Garmin VENU 2<br>Venu 2 -...<br>€222.22<br>Aika Juwelliere...<br>+€5.95 shipping<br>12 days max ·<br>Pedometer, Sle...<br>By Google | <br>Zeus - Smartwatch  ...<br>€89.95<br>StahlGear<br>Free shipping<br>★★★★★ (146)<br>10 days max ·<br>Pedometer, Sle...<br>By Klarna | <br>Gard Pro Health Smartwatch 2+...<br>€79.99<br>Gard Pro Deuts...<br>Free shipping<br>★★★★★ (6k+)<br>Pedometer, Sleep Tracker, Calori...<br>By Google | <br>Zeus - Smartwatch  ...<br>€89.95<br>StahlGear<br>Free shipping<br>★★★★★ (146)<br>Pedometer, Sleep Tracker, Calori...<br>By Klarna |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Places



**Kaufland Passau**  
4,3 ★★★★★ (2,8K) · Supermarket  
Neuburger Str. 128 · 0851 966350  
**Open** · Closes 8 pm  
In-store shopping · [Website](#)



**Rewe**  
4,1 ★★★★★ (1,3K) · Supermarket  
Nibelungenpl. 5 · 0851 4901046  
**Open** · Closes 8 pm  
In-store shopping · [Website](#)



**EDEKA Schwaiberg**  
4,2 ★★★★★ (120) · Supermarket  
Nikolastraße 2 · 0851 20091420  
**Open** · Closes 8 pm  
[Website](#)



# Societal Challenges which require a grip on the Web

## Misinformation / Disinformation

- Monopolistic ranking creates single points of failure for information quality
- Lack of index transparency prevents independent auditing and cross-language disinfo detection
- Open indices enable reproducible fact-checking pipelines

## Defence & Security

- European OSINT depends on US-controlled indices — strategic vulnerability in crisis scenarios
- Adversarial SEO ("search engine poisoning") is a documented hybrid warfare tactic
- No sovereign search infrastructure for independent web monitoring

## Guidance & Algorithmic Bias

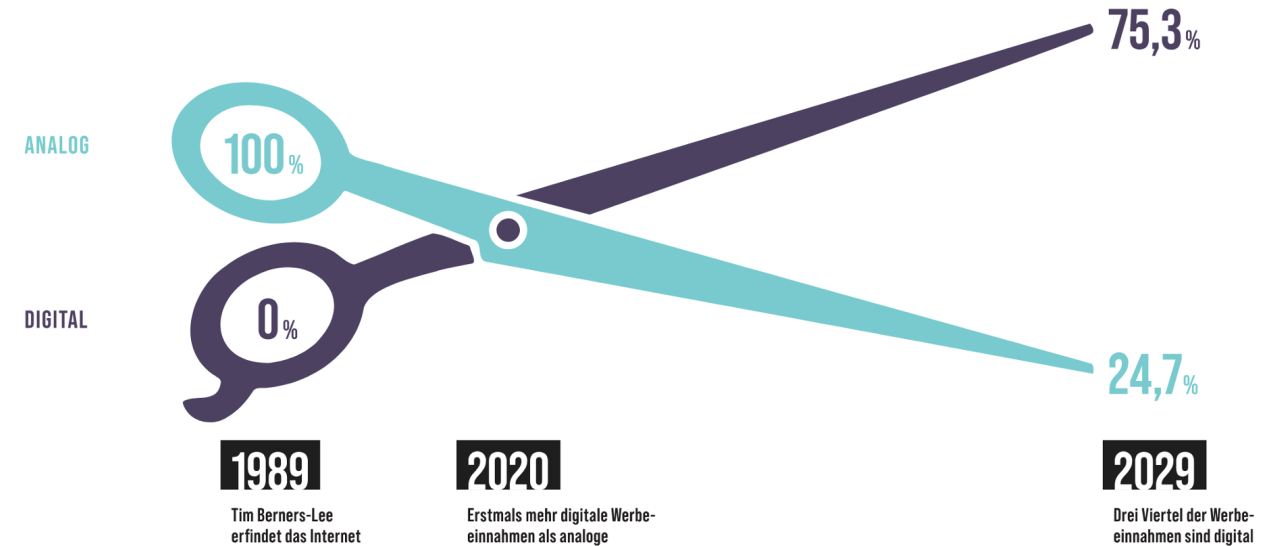
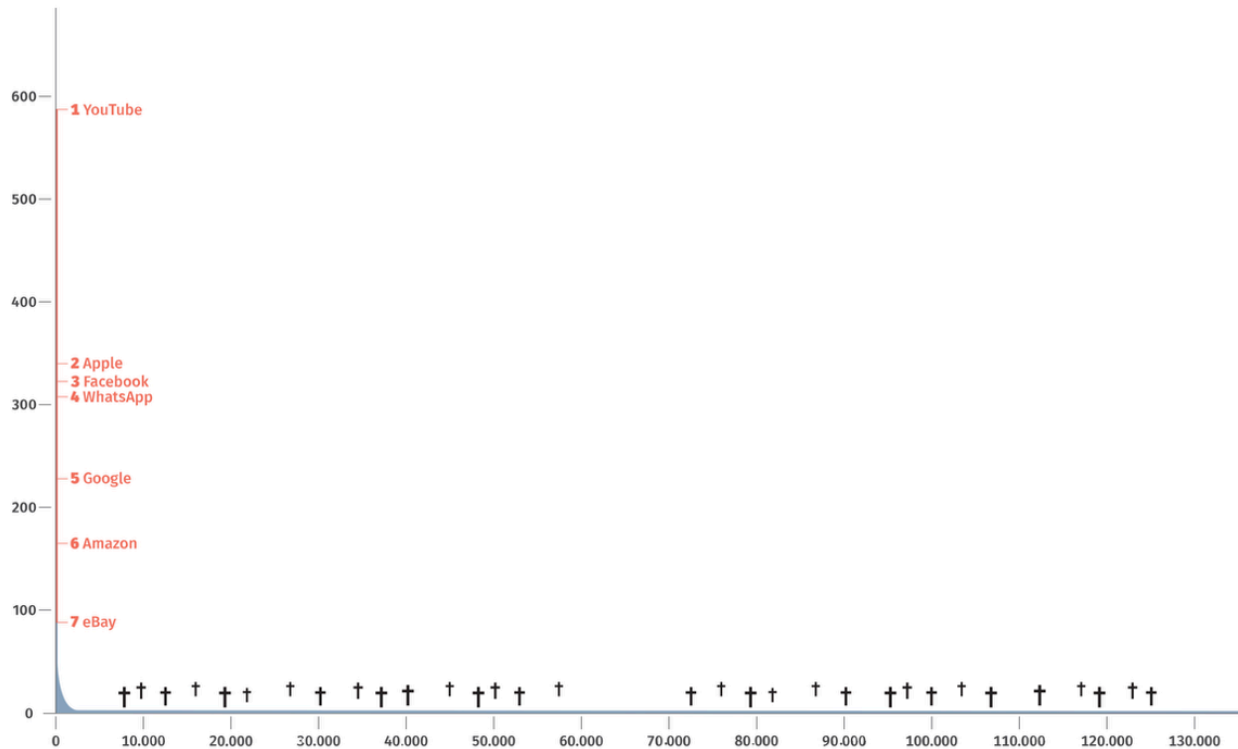
- Search autocompletions and knowledge panels shape health, legal and consumer decisions
- Filter bubbles create asymmetric information through personalization
- Low-income users disproportionately affected by low-quality results

## Digital & Information Literacy

- Most users cannot distinguish ads from organic results — "trust by ranking" effect
- Algorithmic literacy remains low even among university students
- Transparent, inspectable search infrastructure is a prerequisite for teaching information literacy effectively

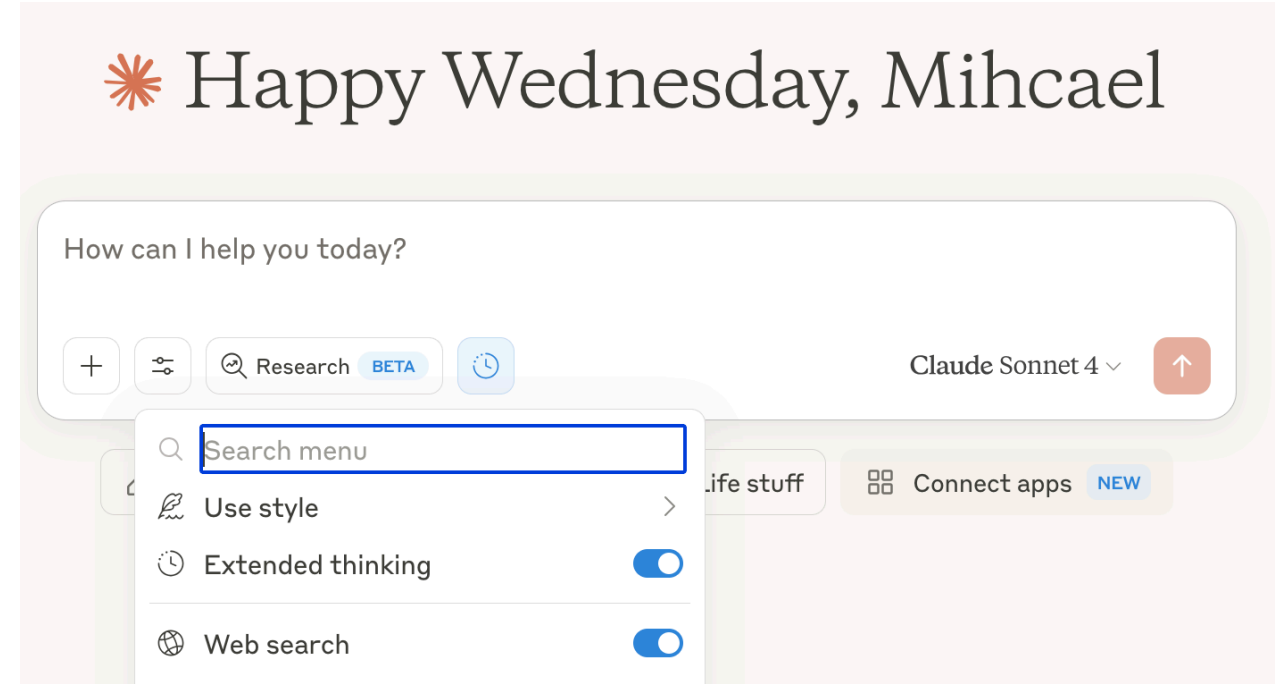
# Europe is not present in the web

- Advertising revenue consolidated by US gatekeepers
- European content, languages, and perspectives underrepresented in search results
- No European alternative for web-scale data access

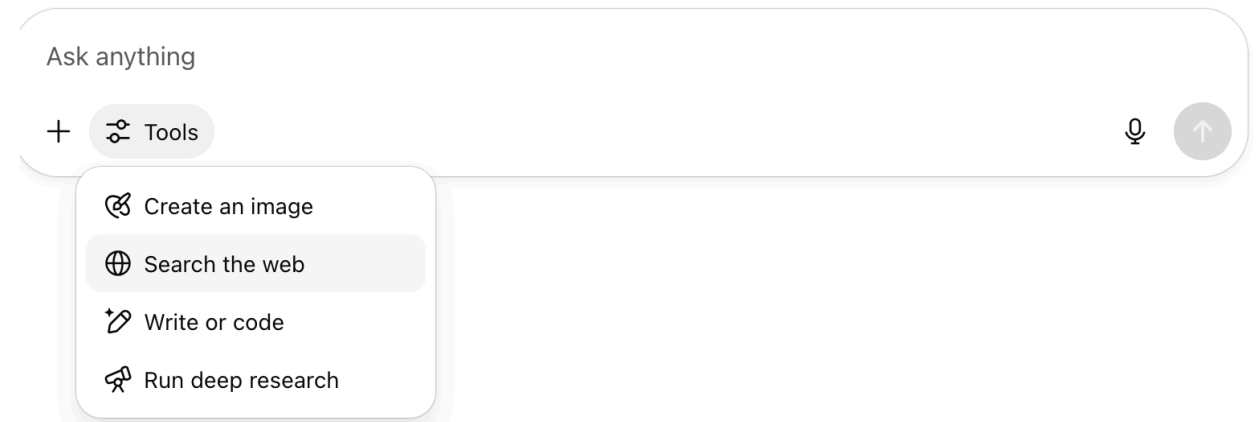


# The web drives innovation and AI

- Web data is essential for training LLMs and VLMs
- Retrieval Augmented Generation (RAG) requires web-scale document collections
- Agentic AI systems rely on web search as a core tool
- Web search remains an integral part of modern AI pipelines



Hey, Michael. Ready to dive in?





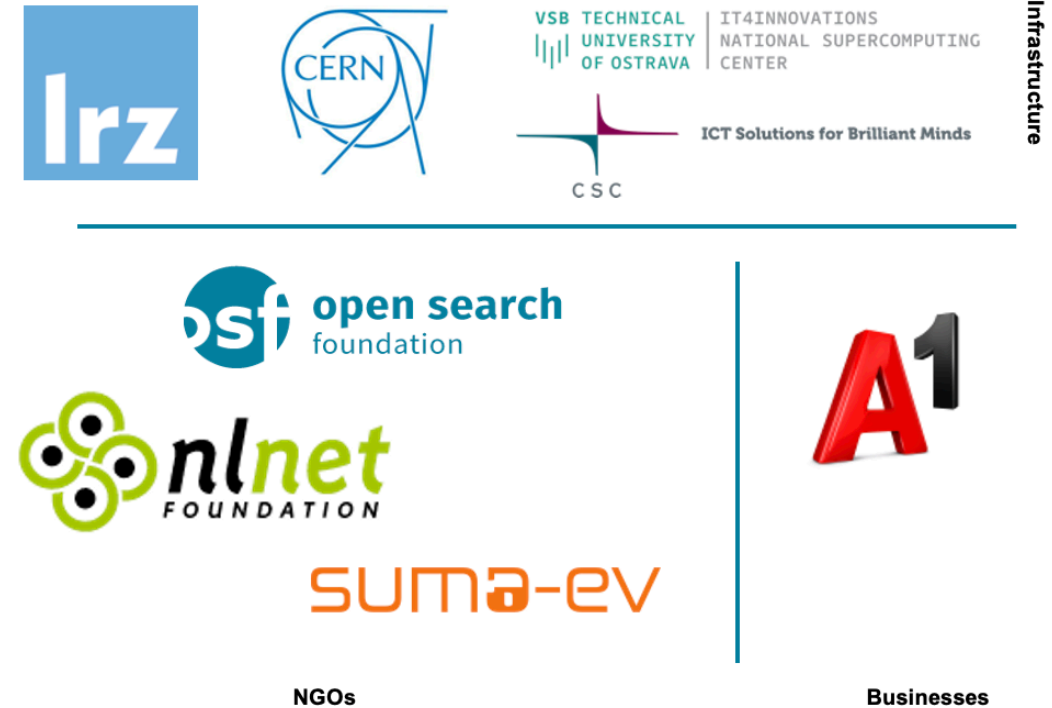
# OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

## Four Objectives

- 1. Open Technology Stack** Open-source pipelines for crawling, preprocessing, indexing
- 2. Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
- 3. Added Value Services** Dashboard, tools, data access APIs
- 4. Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

## 14 Core Partners



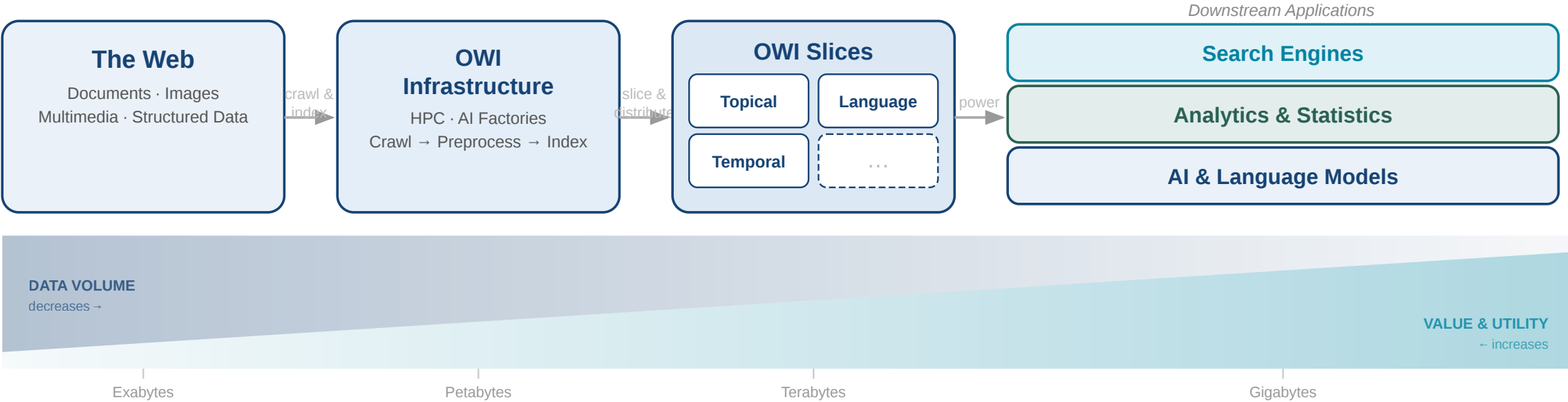
Plus 16 financially supported third-party projects (FSTPss)

Funded by **Horizon Europe** (NGI programme)

# The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

**Our Proposition:** A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe



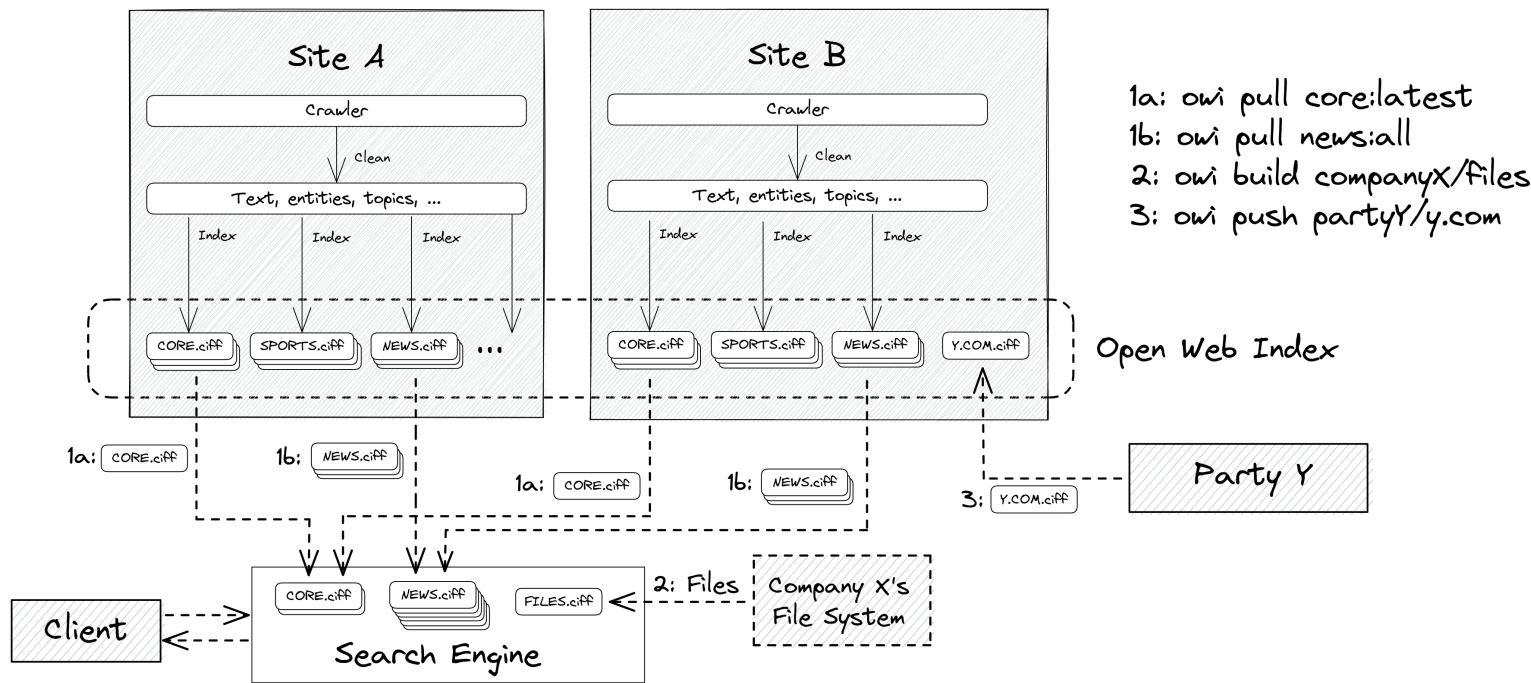
# What is the Open Web Index?

# The Open Web Index

**The Open Web Index** — a federated, daily-published web index built across European HPC sites and consumed as composable shards.

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## Federated production

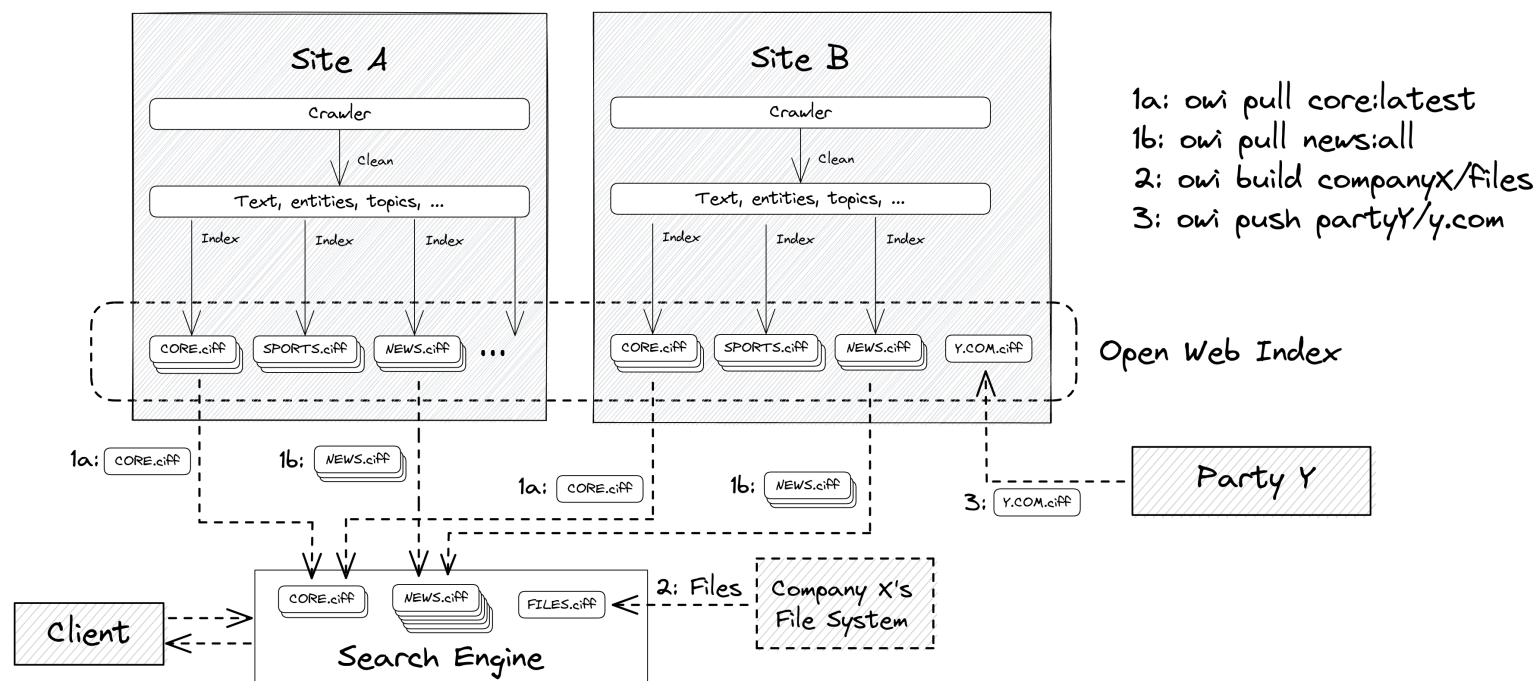
- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

## Three access verbs

- owi pull — download pre-built shards
- owi build — index your own data
- owi push — contribute shards back

# The Open Web Index

**The Open Web Index** — a federated, daily-published web index built across European HPC sites and consumed as composable shards.



## Federated production

- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

## Three access verbs

- owi pull — download pre-built shards
- owi build — index your own data
- owi push — contribute shards back

## Three output types per daily run — the building blocks of a shard:

- index.ciff.gz — sparse inverted index (CIFF format; pre-configured tokenizer)
- metadata\_\*.parquet — document metadata & enrichments (topics, microdata, locations, language, etc.; extensible)
- Dense embeddings (GPU-enriched slice, Jina V3) for AI-based search and classification (on selected subsets)

# OWI at a Glance — Statistics (Feb 2026)

## Crawling

|                 |                           |
|-----------------|---------------------------|
|                 |                           |
| Pages crawled   | ~39 B total, ~12 B unique |
| Unique hosts    | ~500 M (extrapolated)     |
| Crawling effort | 1,927 machine-days        |
| Avg. throughput | ~20 M URLs / day          |

## Index

|                   |                            |
|-------------------|----------------------------|
|                   |                            |
| Documents indexed | ~10.7 B                    |
| Unique domains    | ~75 M                      |
| Public datasets   | 511 main / 1,511 total     |
| Index size        | 45 TB (main) / 50 TB total |
| Embeddings        | 119 M docs · 884 GB        |

**Top languages:** English (38.9%), German (7.2%), Spanish (6.4%), French (5.7%), Russian (5.2%)

**Multimedia Objects:** Roughly 2-3 quality images and 0.5 Videos per Web-Document

**Comparison:** To meet commercial index size, OWI would need to scale by a factor of 20× — factor 10–50× for multimedia

# IN DETAIL

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# Index Format: CIFF + Parquet

## CIFF (Common Index File Format)

A Protobuf schema for sparse inverted files — the OWI standard interchange format:

- **Header** — collection stats: document count, unique terms, avg. document length
- **Postings lists** — per term: document frequency + collection frequency + list of (docid, tf) pairs
- **Document records** — internal numeric ID ↔ external URL + document length

Designed so any existing search engine can import the data and start ranking immediately. Extensions proposed for OWI-specific metadata.

## Parquet Metadata Files

Distributed alongside each CIFF shard — aligned partitions containing:

- URL, title, language, crawl date
- Quality/importance scores from Resilipipe
- Topic / entity enrichments (WP 2)
- Enables SQL-style analytics without downloading the full index

## Dense Embeddings (GPU slice)

- High-quality / high-importance doc subset
- Jina V3 embeddings, 884 GB in public datasets
- Enables semantic search and RAG on OWI content

# Dataset Structure & Versioning



## OWI Dataset

- Year=2023

- month=12

- day=24

-language=eng

index.ciff.gz

metadata\_#.parquet

-language=....

- changelog.json

Each **daily shard** follows a consistent folder hierarchy:

### OWI Dataset

```
└─ year=YYYY
   └─ month=MM
      └─ day=DD
         ├── language=eng
         │   ├── index.ciff.gz
         │   └── metadata_*.parquet
         ├── language=deu
         │   └── ...
         └─ changelog.json
```

- One dataset per data center per day
- Collections: main, curlie\_full, gpu (embeddings), and domain-specific slices
- changelog.json tracks document additions and deletions (take-down compliance)
- Git-like versioning via **owilix**: pull, push, build

# Language Coverage

## Top 15 Languages (main collection)

| Rank | Language | Share |
|------|----------|-------|
| 1    | English  | 38.9% |
| 2    | German   | 7.2%  |
| 3    | Spanish  | 6.4%  |
| 4    | French   | 5.7%  |
| 5    | Russian  | 5.2%  |
| 6    | Japanese | 4.9%  |
| 7    | Chinese  | 3.5%  |
| 8    | Italian  | 2.8%  |
| 9    | Polish   | 1.6%  |

## Collection Breakdown

| Collection  | Datasets | Size   |
|-------------|----------|--------|
| main        | 511      | 45 TB  |
| total (all) | 1,511    | 50 TB  |
| embeddings  | 58       | 884 GB |

**Dedicated language slices** — individual CIFF shards per language enable language-specific downstream applications without full-index downloads

All data, software and formats published as **open source** under permissive licences · Archived on Zenodo



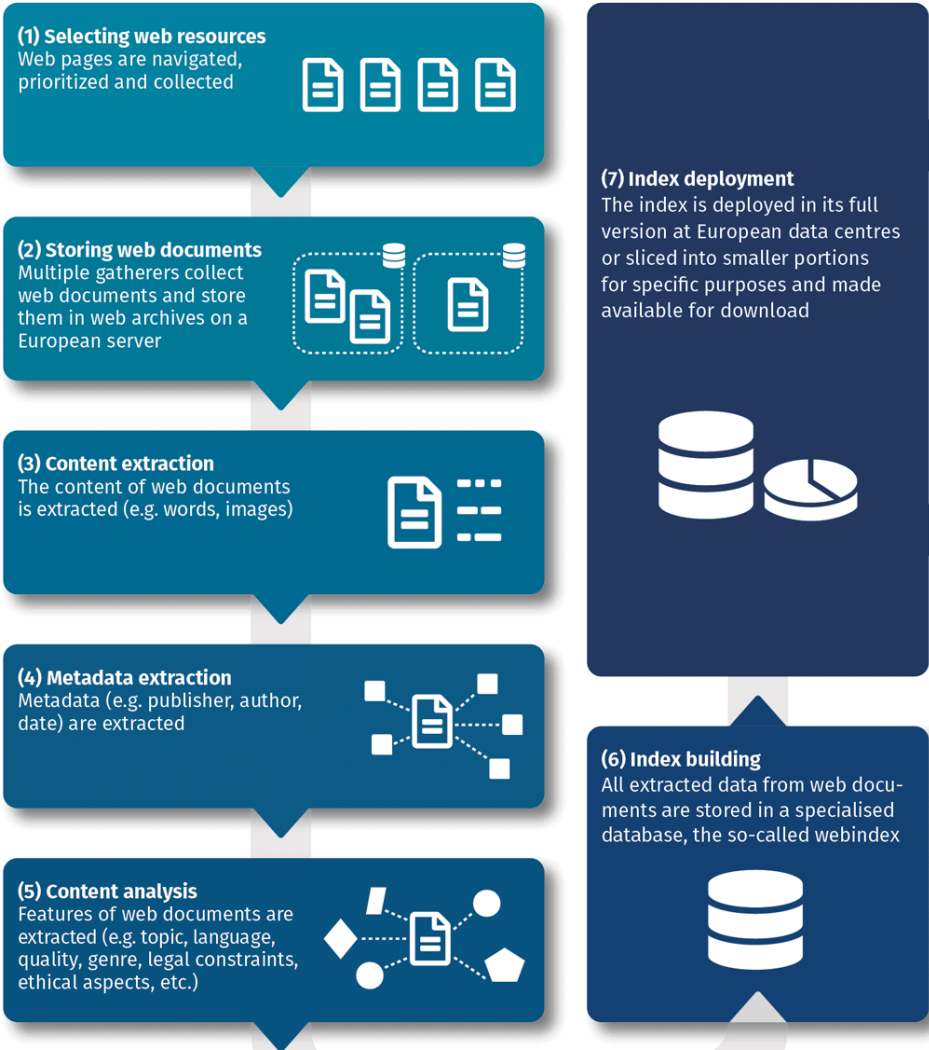
| Rank | Language | Share |
|------|----------|-------|
| 10   | Dutch    | 1.5%  |

# Building the Open Web Index

# Conceptual Workflow

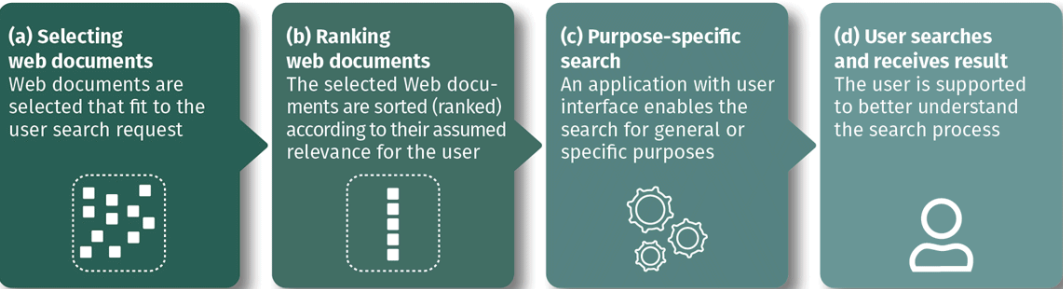
## Index Generation

Web resources are selected and retrieved, their content and metadata are analysed, and all data stored in the index database.



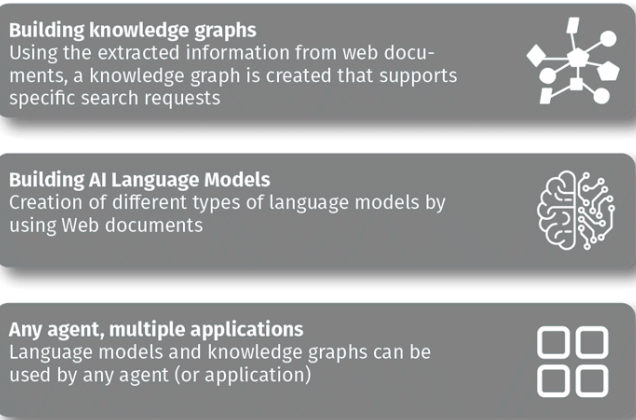
## Search Applications

A user search request will be answered by a search application that makes use of the open web index.



## Data Products

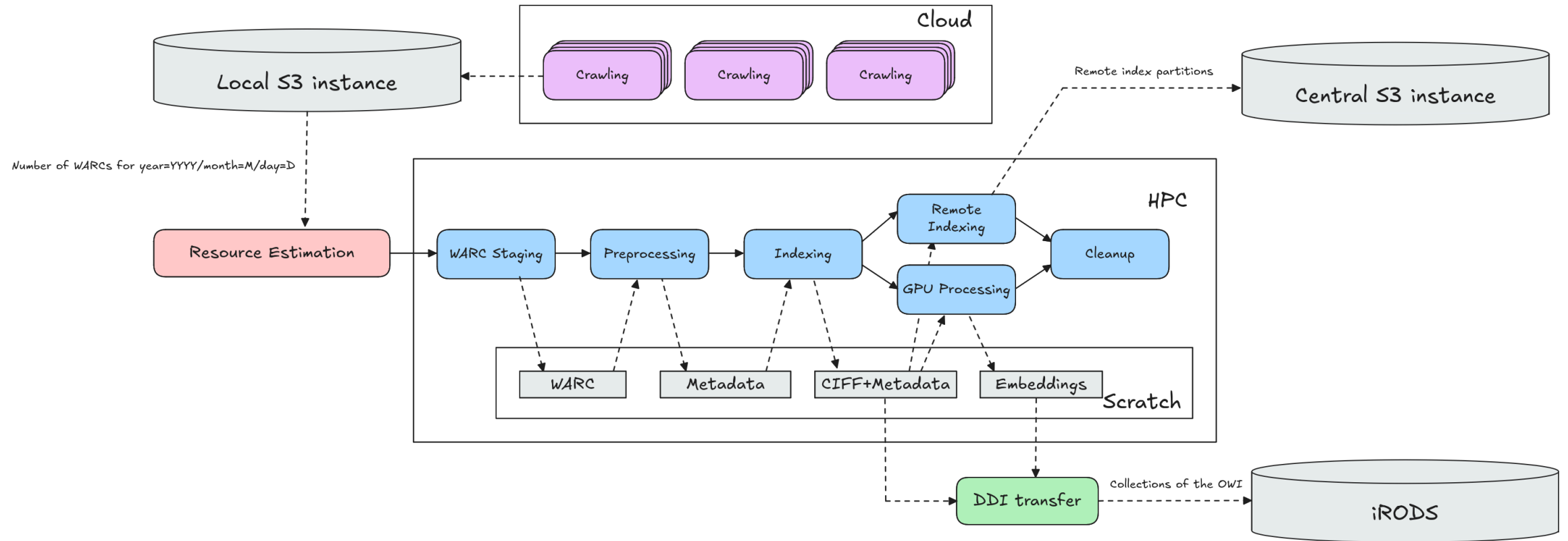
Knowledge representation models will be created using the open web index, in order to be used by any agent and for many applications



...



# Daily Workflow Pipeline



## Crawlig Infrastructure (2026)

- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

## Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement

# IN DETAIL

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# The OpenWebSearch.eu Pipeline — Overview

## Step 1 — Crawling

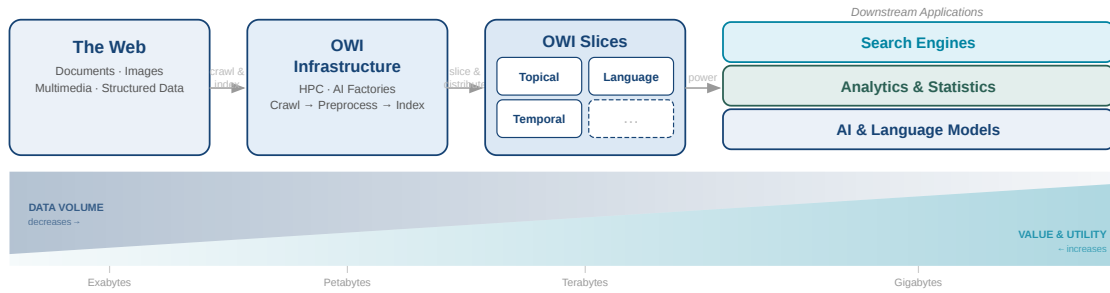
- 100 M URLs / day, 25 machines in 3–4 DCs
- European domain focus
- Respects robots.txt, Gen AI opt-out
- Full transparency via Dashboard

## Step 2 — Preprocessing (Resilipipe)

- HTML → clean text (ratio 1:5)
- Metadata: JSON-LD, Microdata, OpenGraph
- Semantic enrichment: language, topics, NER, geo-locations
- Structured data from 60 % of pages

## Step 3 — Index & Distribution

- Sparse full-text index (Cliff / Lucene)
- Feature index: link structure, style, genre
- Dense index: Jina V3 embeddings
- 810+ daily datasets via Owilix / Dashboard



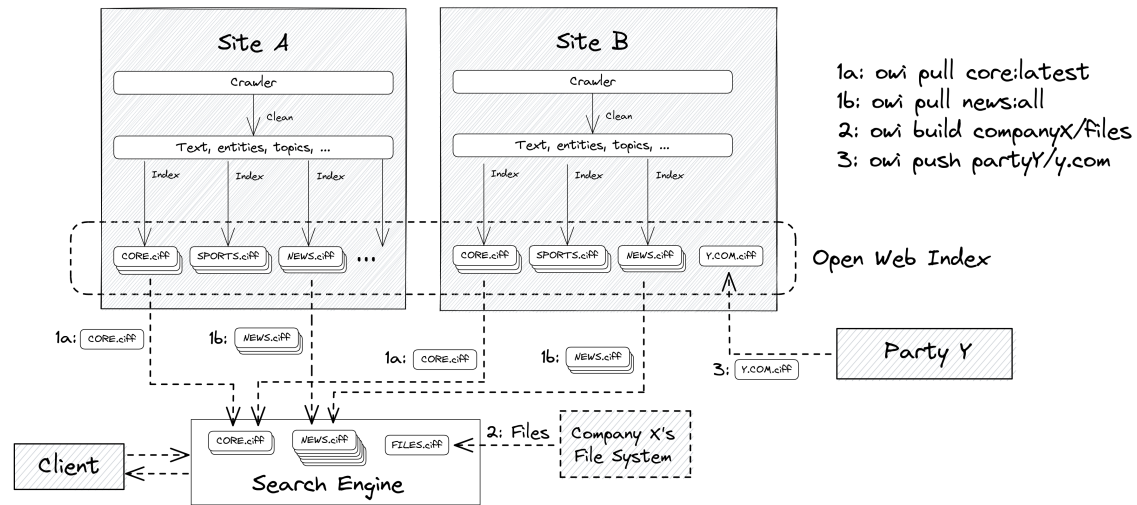
## Scale (as of early 2026)

- **25 VMs** across 3–5 data centres (IT4I, CSC, LRZ, CERN) running 24/7
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index in **810+** public datasets
- Factor **20 cheaper** than commercial indices

## Architecture principles

- Federated storage (iRODS + S3), cross-DC workflow via LEXIS
- Client-side SQL access via Owilix CLI
- Open format: CIFF sparse index + Parquet files
- Dense embeddings (Jina V3) available on select slices

# Architecture & Scale



## Four Cluster Tiers per Data Center

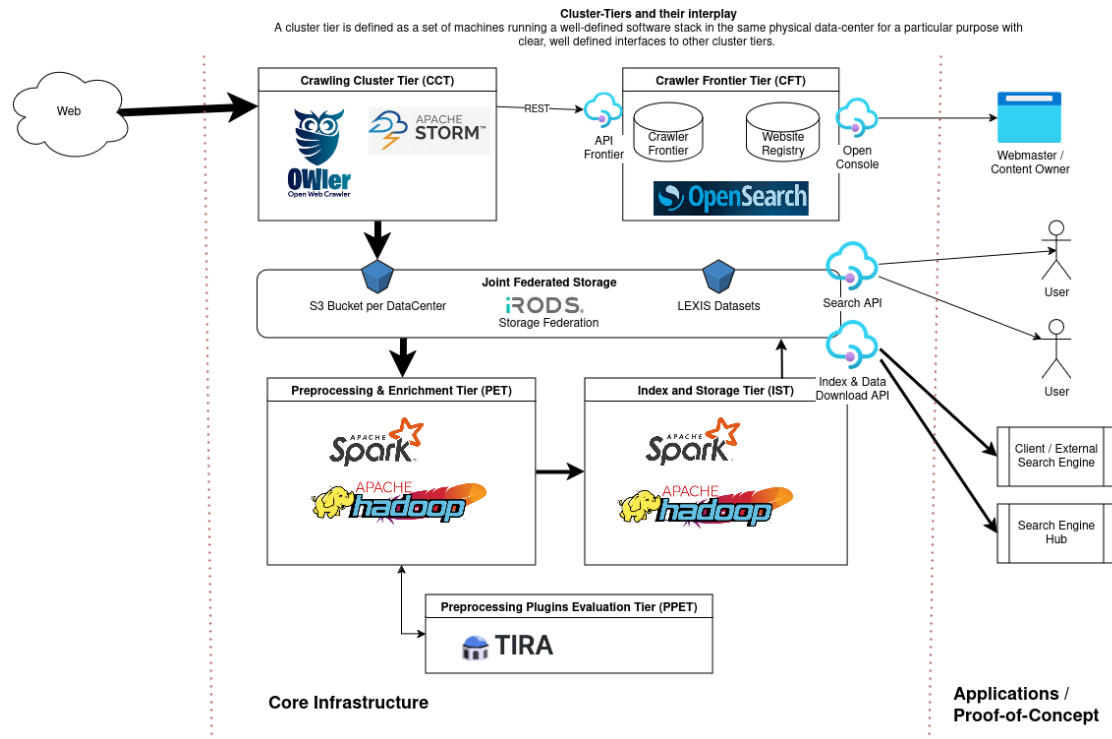
- **CCT** — OWLer crawler (StormCrawler-based); produces WARC files into S3
- **CFT** — singleton ScyllaDB URL frontier at CSC; coordinates all crawl sites
- **PET** — Resilipipe (Python multiprocessing + Resiliparse): WARC → Parquet
- **IST** — Open Web Indexer (Apache Spark): Parquet → CIFF + remote index

**Deployed at:** IT4I@VSB · BADW-LRZ · CSC · CERN + FSTP partners

## Scale (early 2026)

- **25 VMs** across 3–5 data centers running 24/7
- **100 M URLs / day** crawling · **~1.1 PB** raw data
- **~50 TB** index · **810+ public datasets**

# Cluster Tiers in Detail



## Crawling (WP 1)

- **OWler** (distributed StormCrawler): multi-site web crawl, WARC output
- Peak: ~100 M URLs/day at 4 partner sites (UNI PASSAU, LRZ, IT4I, CSC + EXAION)
- **ScyllaDB** frontier: dedup, politeness, cross-DC coordination

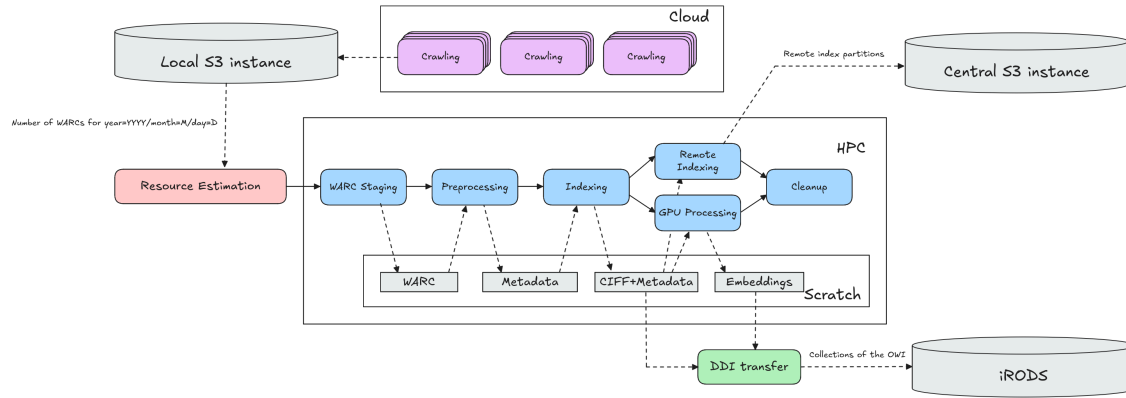
## Preprocessing (WP 2)

- **Resilipipe** on Spark: HTML → clean text, metadata extraction, topic/entity tagging
- **PPET** (singleton): plugin evaluation via TIRA before promotion to production
- GPU tier: quality-filtered docs → Jina V3 dense embeddings

## Indexing (WP 3)

- **Open Web Indexer** on Spark: Parquet → CIFF shards per collection + language

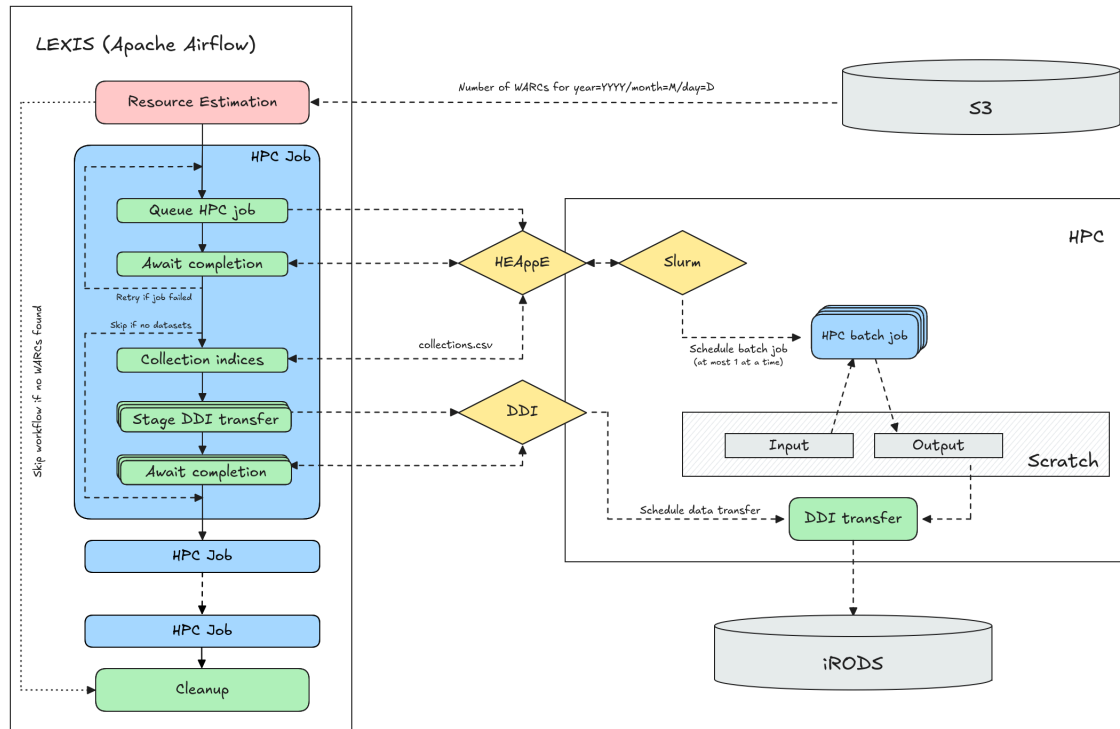
- Also produces **remote index** (DuckDB/OUR<sup>2</sup>S-queryable Parquet partitions)
- Federated storage: **iRODS** + S3 via LEXIS DDI layer



Each day at **03:00 AM** the pipeline triggers:

1. **WARC Staging** — copy new WARC files from S3 to HPC scratch
2. **Resilipipe** — WARC → gzipped Parquet (text, metadata, enrichments)
3. **Open Web Indexer** — Parquet → CIFF shards per collection / language
4. **GPU Processing** — Resilipipe GPU → Jina V3 embeddings for quality slice
5. **Remote Index** — Parquet partitions for DuckDB / OUR<sup>2</sup>S remote queries
6. **DDI Transfer** — push output collections to iRODS via LEXIS DDI layer
7. **Cleanup** — remove intermediate HPC scratch data

All intermediate data stored on **HPC scratch** between steps — restartable on partial failure.



**LEXIS** (Apache Airflow) manages the full daily pipeline:

- **Resource estimation:** counts WARC's for that date → computes Slurm allocation
- **HEAppE:** submits and monitors HPC batch jobs at each data center
- **DDI layer:** transfers output collections to iRODS after each job succeeds
- Automatic retry on failure; session tokens refreshed throughout

## Workflow configuration (JSON per DC):

- `crawl_location / hpc_location / storage_location` — Airflow Connections
- Per-task: `resources`, `max_retries`, `stage_ddi`, `depends_on`
- Dev / test / prod variants with reduced resource profiles

**Data center connections:** S3 (crawl), HEAppE (HPC scheduling), iRODS (persistent storage) — fully configurable per data center.

## Index & Federation

- ~**1.1 PB** raw data · ~**50 TB** index data · **810+** public datasets
- **10–20 TB** new index data per month
- Approx. **factor 20** smaller than commercial indices (factor 10–50× for multimedia)

## Federated Architecture

- Distributed over **IT4I, CSC, LRZ, CERN** — onboarding further AI Factories
- Cross-DC workflow execution via LEXIS platform
- Federated storage: iRODS + S3; client-side SQL access via owilix / DuckDB
- Dense embeddings available on select slices

# Step 1: Crawling the Web

# Step 1: Crawling the Web



## OWLer — Open Web Crawler

- Up to **105 M URLs / day** across 4 European data centers
- Central URL Frontier at **CERN** — 2 B+ URLs in crawl space
- **1.25 PiB** collected · **1,927 machine-days** · ~39.6 B fetches
- European language and domain bias
- Only HTML; No headless browsing

## Constraints

- Downstream processing caps: **1–2 TB/day per DC**
- Factor ~20× below commercial freshness
- Requires 2–4 dedicated engineers for 24/7 ops

# IN DETAIL

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# OWLer — Three-Tier Crawling Architecture

## Tier 1 — Crawling Cluster (StormCrawler)

- StormCrawler on Apache Storm — distributed, fault-tolerant, linearly scalable
- VMs at **LRZ**, **IT4I**, **CSC** (OpenStack)
- Three pipeline modes:
  - **Regular** — explorative hyperlink-following (highest coverage, highest risk)
  - **Sitemap** — sitemap-driven discovery (lower risk, curated scope)
  - **WARC2WARC** — ingest external WARCs (e.g. Common Crawl) without live fetching
- Outputs WARC files → fed into downstream preprocessing pipeline

## Tier 2 — URL Frontier (CERN)

- Central gRPC service coordinating all crawlers
- Manages 2 B+ URL crawl space in OpenSearch
- Stream-based and batch-based APIs
- Politeness: delay between fetches per domain

## Tier 3 — Persistence

- OpenSearch backend for URL state and logs
- Apache Flink for real-time log aggregation
- Published as OWI Dashboard datasets

# URL Frontier — Coordination & Scheduling

## Centralized URL Management

- Single authoritative component — simplifies monitoring, debugging, coordination
- gRPC API with Protocol Buffers — language- and crawler-neutral
- Benchmark: **295.9 M simulated fetches in 12 h**, 2 B URL crawl space, 80 GB RAM peak
- Up to **5,500 URLs/second** (batch mode), 2,000/s (stream mode)

## Scheduling Strategy

- Each URL gets a nextFetchDate — continuous, not round-based
- Freshness via If-Modified-Since + MD5 content hash comparison
- **Quality lists** to prioritize high-value content:
  - ClueWeb22 Super Head (Category B)
  - Curlie / DMOZ directory (human-curated)
  - Wikipedia external link outlinks

### URL Tagging (*% of crawl space*)

| Tag               | %      |
|-------------------|--------|
| HTML              | 83.1 % |
| GenAI allowed     | 79.6 % |
| Index allowed     | 77.0 % |
| Wikipedia outlink | 48.7 % |
| SuperHead ClueWeb | 9.66 % |
| Curlie category   | ~3.7 % |
| Blacklisted       | 0.18 % |

# Crawl Statistics — Three Operational Phases

Three major crawl campaigns across 120 weeks (Nov 2023 – Feb 2026) at European HPC centres

| Phase                 | Machine-days | Data (TiB) | Fetches (B) | Hosts (M) | Unique pages | Recrawl ratio |
|-----------------------|--------------|------------|-------------|-----------|--------------|---------------|
| Aug 2023 – Aug 2024   | 760          | 315        | 10.76       | 40.5      | 1.17 B       | 9.2×          |
| Apr 2025 – Jul 2025   | 189          | 175        | 4.76        | 162.0     | 3.76 B       | 1.2×          |
| Dec 2025 (validation) | —            | —          | —           | 20.5      | 341 M        | —             |
| Total                 | 1,927        | ~1,280     | ~39.6       | —         | ~12.6 B      | —             |

**Peak throughput**  
~105 M URLs/day  
88% fetch success rate  
184 languages detected

**Practical cap**  
1–2 TB/day/DC  
(downstream processing limit)  
~20× below commercial freshness

**To reach production scale**  
2–4 dedicated engineers  
24/7 continuous operation  
~50× more bandwidth + storage

# Auxiliary Crawls — Targeted Fill-In Collections

Beyond general-purpose web crawling, specialized scrapers harvest structured sources that standard hyperlink-following cannot reach:

| Source                     | Coverage                                              | Volume                                              |
|----------------------------|-------------------------------------------------------|-----------------------------------------------------|
| Wikipedia                  | 10 languages (en, de, fr, es, it, nl, pl, sv, ar, ja) | 83 GB WARC/month · 6.5 M external domains extracted |
| Mastodon                   | 150 instances (top 1,000 by activity)                 | 5 GB/day · 150 GB/month                             |
| OpenAlex                   | Scientific publications snapshot + incremental API    | 117 GB compressed WARC                              |
| Pangaea / OpenAire / Eudat | Scientific data repositories                          | via API → WARC                                      |
| Internal GitLab            | Code repositories                                     | via API → WARC                                      |

All auxiliary crawlers produce **WARC files** and feed into the same downstream OWI pipeline as the general-purpose crawler.

## Why auxiliary crawls?

- Wikipedia outlinks seed high-quality URLs into the general-purpose frontier
- Scientific publications enable Open Science Search use cases
- Mastodon adds social-web content beyond the crawlable public web
- Complements gaps in general-purpose coverage (API-only content, structured dumps)

# Compliance, Transparency & Webmaster Control

## Ethical Crawling Practices

- Respects **robots.txt**, X-Robots-Tag, noindex directives
- Two derived binary flags stored in every Parquet record:
  - Index — allowed for web indexing / search
  - GenAI — allowed for generative AI model training

| Index | GenAI | Share  |
|-------|-------|--------|
| ✓     | ✓     | 76.2 % |
| ✓     | ✗     | 0.74 % |
| ✗     | ✓     | 3.34 % |
| ✗     | ✗     | 19.7 % |

- **Demarcated crawling subnets** — IP ranges separated from institutional access to avoid license infringements
- **Blacklisting**: Ultimate Hosts Blacklist + Université Toulouse Capitole lists (0.18% of URLs blacklisted)

## Open Webmaster Console

- Any webmaster can look up their domain in the OWI
- **Domain verification** to unlock management
- **Takedown requests** at URL or subdomain level → propagated to all index users
- **Crawl constraint config** via OWLer wizard
- Ethical self-assessment survey required before dataset download

Incident log: Bank of America (robots.txt ambiguity), ACM Digital Library (licensed content), anonymous platform (forum spam false positive) — all investigated and resolved

# Step 2: Preprocessing

## Step 2: Preprocessing

### Resilipipe — WARC → Parquet Pipeline

- Two-stage pipeline: **Resilipipe (CPU)** for extraction & enrichment, **Resilipipe GPU** for embeddings & classifiers
- Modular architecture — 10+ pluggable modules (links, structured data, quality scoring, trigger warnings, ...)
- Custom parallelization on HPC: SLURM + Apache Airflow orchestration
- Output: column-oriented **Apache Parquet** with per-document metadata, embeddings, and prediction scores

### Scale & Constraints

- **108 M documents** embedded with jina-embeddings-v3 (785 GB across 55 published datasets)
- GPU stage bottleneck: embedding throughput limits end-to-end pipeline speed
- Downstream cap of **1–2 TB/day per data center** inherited from crawling

### Key Enrichments

- Content extraction & language detection
- Link graph & structured data (FAQs, geo, licenses)
- Curlie taxonomy labels
- Quality scoring (ONNX)
- Trigger warnings (14 languages, 19 categories)
- Dense embeddings (jina-embeddings-v3)
- Phishing classifier & LLM detector

# IN DETAIL

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# Resilipipe (CPU) — Modular WARC → Parquet Pipeline

## Architecture

- Input: **WARC files** from OWLer crawl → Output: **Apache Parquet** (columnar, query-friendly)
- Custom parallelization replacing PySpark:
- **ProcessWorkers** → **ParsingThreads** → **WritingThread**
- Three module types:
- **BasicModule** — operates on raw WARC records (headers, links)
- **HTMLModule** — receives parsed HTML DOM tree (structured data, quality)
- **PostProcessingModule** — runs after all others (collection indices, forum classification)

## CPU Modules

| Module               | Function                                  |
|----------------------|-------------------------------------------|
| ows_headers          | HTTP headers & metadata                   |
| links                | Link graph extraction                     |
| curlielabels         | Curlie taxonomy classification            |
| structured_data      | FAQs, licenses, geo, opening hours, phone |
| pyterrier_quality    | Quality scoring (ONNX)                    |
| trigger_warnings     | Content warnings (14 langs × 19 types)    |
| collection_indices   | Per-collection index metadata             |
| forum_classification | Forum / discussion detection              |

# Resilipipe GPU — Embeddings & Prediction Heads

## Embedding Pipeline

- Model: **jina-embeddings-v3** — multilingual, 8192-token context, task-specific LoRA adapters
- Selected from evaluation on 3.3 GB multilingual OWI test dataset
- Chunking via embedding model's tokenizer — embeddings computed **per chunk** (supports RAG retrieval)
- **55 published datasets** containing embeddings for **108 M documents** (785 GB total)
- Research track: embedding compression to reduce storage footprint

## GPU Parallelization

- ReadingWorker → EmbeddingWorker (per GPU) → AggregationWorker → ModuleWorker (per GPU) → WritingWorker
- Prediction heads: small `torch.nn.Module` networks trained on chunked embeddings
- Each GPU processes batches independently; aggregation worker merges results

## Prediction Heads

| Head                | Task                         |
|---------------------|------------------------------|
| Phishing classifier | Detect phishing / scam pages |
| LLM detector        | Flag LLM-generated content   |

## Why jina-embeddings-v3?

- Best multilingual performance on OWI evaluation
- Task-specific adapters (retrieval, classification, clustering)
- Long-context support (8192 tokens)
- Open-weight model

# Computational Ethics — Content Safety & Detection

## Trigger Warning System

- Lexical scoring approach across **14 languages** and **19 warning categories**
- Categories include: violence, self-harm, sexual content, hate speech, substance abuse, eating disorders, and more
- Per-document scores stored in Parquet output — downstream consumers can filter at chosen thresholds
- Designed for transparency: users of OWI data can apply their own policies

## LLM-Generated Content Detection

- Native advertising detection research (CLEF 2025 shared task)
- LLM detector prediction head identifies machine-generated text
- Critical for data quality: prevents AI-generated content from polluting training corpora and search indices

## Why this matters

- Web-scale data inevitably contains harmful content
- Index consumers (search engines, researchers, AI trainers) need **per-document signals** to make informed filtering decisions
- OWI provides the signals — **not** the filtering policy

## Design Principle

Signal, don't censor — OWI enriches data with safety metadata; downstream users decide thresholds and policies.

# HPC Deployment — SLURM + Apache Airflow

## Orchestration Stack

- **Apache Airflow** manages DAG workflows across both Resilipipe stages
- **SLURM** scheduler allocates CPU and GPU nodes at HPC data centers
- Automated WARC staging: **S3** → **scratch filesystem** before processing

## Workflow Stages

1. **WARC Staging** — pull WARC files from S3/iRoDS to local scratch
2. **Resilipipe (CPU)** — extract, enrich, write Parquet
3. **Resilipipe GPU** — compute embeddings, run prediction heads, write enriched Parquet
4. **Publication** — upload enriched datasets to OWI distribution (LEXIS / S3)

## Deployment at Scale

- Runs at **LRZ, IT4I, CSC** HPC centres
- CPU and GPU jobs submitted as SLURM batch arrays
- Airflow DAGs monitor progress, handle retries, track dataset lineage

## End-to-End Flow

```
WARC (crawl)
  → Resilipipe CPU
    → Parquet (enriched)
      → Resilipipe GPU
        → Parquet (+ embeddings)
          → OWI distribution
```

# Step 3: Building the Index

# Index Format: CIFF + Parquet

## CIFF (Common Index File Format)

A Protobuf schema for sparse inverted files — the OWI standard interchange format:

- **Header** — collection stats: document count, unique terms, avg. document length
- **Postings lists** — per term: document frequency + collection frequency + (docid, tf) pairs
- **Document records** — internal numeric ID ↔ external URL + document length

Designed so any existing search engine can import the data and start ranking immediately.

## Parquet Metadata Files

Distributed alongside each CIFF shard — aligned partitions containing:

- URL, title, language, crawl date
- Quality / importance scores from Resilipipe
- Topic / entity enrichments (WP 2)
- Enables SQL-style analytics without downloading the full index

## Dense Embeddings (GPU slice)

- High-quality / high-importance doc subset
- Jina V3 embeddings, 884 GB in public datasets
- Enables semantic search and RAG on OWI content

# Dataset Structure & Versioning

Each **daily shard** follows a consistent folder hierarchy:

```
OWI Dataset
├── year=YYYY
│   ├── month=MM
│   │   ├── day=DD
│   │   │   ├── language=eng
│   │   │   │   ├── index.ciff.gz
│   │   │   │   └── metadata_*.parquet
│   │   │   ├── language=deu
│   │   │   └── ...
│   │   └── changelog.json
```

- One dataset per data center per day
- Collections: main, curlie\_full, gpu (embeddings), and domain-specific slices
- `changelog.json` tracks document additions and deletions (take-down compliance)
- Git-like versioning via **owilix**: pull, push, build

OWI Dataset

```
- Year=2023
  - month=12

    - day=24

      -language=eng

        index.ciff.gz
        metadata_#.parquet

      -language=....

- changelog.json
```

# Accessing the Open Web Index

# The Dashboard — openwebindex.eu

OWI

Dashboard

SERVICES

Websites

Search

Serci Search EXTERNAL

Crawling

EXPERIMENTAL

Deep Research

UTILITIES

Parse Preview

Ethics Readiness Check

Impressum

Privacy Statement

Terms of Use

Open Web Search Book

OpenWebSearch

Open Web Index

Overview OWI Statistics OWI Datasets OWI Corpora OWI Models

## The Open Web Index

Your daily dose of web data!

Welcome to the Open Web Index (OWI)- your way to slice and dice the Petabytes in the Web to your needs. Our mission is to contribute to a fair, open, diverse and free Web by providing an open Index of the Web for you to use. This dashboard gives an overview over this [Open Web Index](#) and provides some prototype services on top of the OWI.



## What can you find here?

Discover our comprehensive suite of tools and services designed to help you explore and analyze web data.



**OWI Statistics**  
Get an overview of the data we offer



**OWI Datasets**  
Browse through available datasets



**OUR Search**  
Search through URLs and Titles



**Documentation**  
Access tutorials and detailed documentation

**Expect Bugs:** While we try to provide as many valuable datasets and services as possible, please always remember that this is still a research project. So we cannot guarantee that everything works perfectly fine and we cannot take any responsibility if something is not working as intended. Bugs are to be expected, but if you encounter some, please get in touch with us so that we can improve.

Central entry point for the Open Web Index [openwebindex.eu](https://openwebindex.eu)

- **OWI Statistics** — crawl volume, index size, language distribution, dataset timeline
- **OWI Datasets** — browse, filter and download index shards (by DC, date, format, language)
- **Search** — demo content search + SERCI integration

424 registered users · 17,000+ unique visitors · 70,000+ page views

# Dashboard — Statistics & Datasets



- ### OWI Statistics (04/26)
- **1,389 TB** crawled · **281** WARC datasets · **1,696** public datasets · **35.09 TB** index
  - Language distribution chart (English 38%, German 8%, French 6%, ...)
  - Crawl volume over time per data center + dataset collection distribution by type
  - Datasets over time

# Dashboard — Statistics & Datasets

### Available Datasets

The data shows the datasets available for download (i.e. public owi datasets) or upon request (mostly project private warc datasets) by using our [LEXIS Platform](#) or our [OWI Command Line Tool](#). A dataset is a temporal slice, most often a single day, of crawled (warc), preprocessed and indexed data (owi).

All systems are online

All components are functioning properly. Data repositories and services are accessible.

Last Updated: 12/04/26, 15:36

| SYSTEMS | DESCRIPTION | COMPONENTS |            |         | STATUS |        |
|---------|-------------|------------|------------|---------|--------|--------|
| IT4I    | IT4I        | connected  | Repository | Project | Public | ONLINE |
| LRZ     | LRZ         | connected  | Repository | Project | Public | ONLINE |

Filter datasets...

All Collections

All Datacenters

All Resources

Public

Sort by: Collection Date

OWI - Open Web Index

05/04/2026 → 05/04/2026

LEGAL IT4I PUBLIC

204.5 MB

253 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fcb68e42-3191-11f1-8cbe-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

MAIN IT4I PUBLIC

24.3 GB

1,065 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=faba586e-3191-11f1-9f25-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

CURLIE\_FULL IT4I PUBLIC

2.1 GB

365 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fabf3850-3191-11f1-af7e-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

LICENSES IT4I PUBLIC

63.6 MB

163 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fac99de2-3191-11f1-b11f-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

GPU IT4I PUBLIC

3.6 GB

374 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=faf69d860-3191-11f1-9f25-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

GPU IT4I PUBLIC

40.8 MB

6 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

owilix remote pull all/internalID=2f8f678c-2fd5-11f1-8ad8-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

LICENSES IT4I PUBLIC

275.8 KB

2 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

CURLIE\_FULL IT4I PUBLIC

5.3 MB

8 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

MAIN IT4I PUBLIC

108.4 MB

73 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

## OWI Datasets Tab

- Status of repositories
- Filter by datacenter, format (WARC / Parquet / CIFF / Embeddings), language, date
- Dataset cards with download links — LEXIS Portal or owilix CLI
- Parquet file preview directly in the browser

55

# Download Tool: owilix

## owilix — The OWI Command-Line Tool

Git-like versioning for index datasets — pull, build, push:

```
owilix remote pull all:latest/collectionName=main
owilix local ls all:latest
```

Slice by: **domain**, **language**, **topic** (Curly), **sitelist**, **structured data**, **outlinks**  
Authentication via B2ACCESS / Keycloak (LEXIS SSO)

## Remote Index — Query first, then Download

- Parquet partitions directly queryable via **DuckDB** or **Apache Spark**
- Full-text search
- SQL access: filter by language, domain, date — no full-shard download needed
- Enables search + filtering + download (but still in alpha phase)

## OWILIX - Open Web Index CLI

version 5.3.1

python 3.11+

license Apache 2.0

A command-line tool for managing [OpenWebIndex](#) datasets across local and remote repositories.

### Quick Start

#### One-Line Install (Recommended)

```
curl -fsSL https://openwebindex.net/owilix/install.sh | sh
```

<https://opencode.it4i.eu/openwebsearcheu-public/owi-cli/>

```
[→ ~ owilix remote search "michael granitzer" --limit 10
Searching remote index: terms=['michael', 'granitzer'] language=eng representation=main_content mode=OR limit=10
Found 10 results in 113.4s
Search results (10 results in 113.4s)
```

| score | date       | url                                                          |
|-------|------------|--------------------------------------------------------------|
| 3.088 | 2025-12-30 | https://easychair.org/smart-program/ICADL2024/person16.html  |
| 3.010 | 2026-02-19 | https://www.wolframalpha.com/input/?i=michael                |
| 3.010 | 2026-01-19 | https://www.michaelmoennich.com/                             |
| 3.010 | 2025-12-08 | https://michaelkoenig.de/                                    |
| 3.010 | 2026-01-05 | http://www.michaelmoerk.dk/                                  |
| 3.010 | 2026-01-22 | https://michaelkoenig.de/                                    |
| 3.010 | 2026-02-18 | https://www.sktthemes.org/forums/users/dedicationpt/replies/ |
| 3.010 | 2026-02-26 | https://www.abeldiaz3.com/tag/michael/                       |
| 3.010 | 2026-02-20 | https://discourse.psychopy.org/u/michael                     |
| 3.010 | 2026-02-27 | https://thedigitalgobag.com/author/michael/                  |



# The OpenWebSearch Book

Comprehensive Documentation at [openwebindex.eu/book](https://openwebindex.eu/book)

Covers the full OWI ecosystem:

- Getting started with owilix
- Building vertical search engines with MOSAIC
- Data formats: CIFF, Parquet, embeddings
- Pipeline architecture and HPC deployment
- Legal & ethical guidelines for index users

Published as an open, continuously updated online resource — aligned with deliverables and community feedback.

# Access Modes at a Glance

| Tool                      | Mode           | Best for               |
|---------------------------|----------------|------------------------|
| <a href="#">Dashboard</a> | Browser        | Explore, monitor       |
| <a href="#">owilix</a>    | CLI            | Download, slice, build |
| Remote index              | DuckDB/SQL     | Query + download       |
| <a href="#">MosaicRAG</a> | Conversational | RAG & chat over OWI    |

All software, data formats and access tools are **open source**

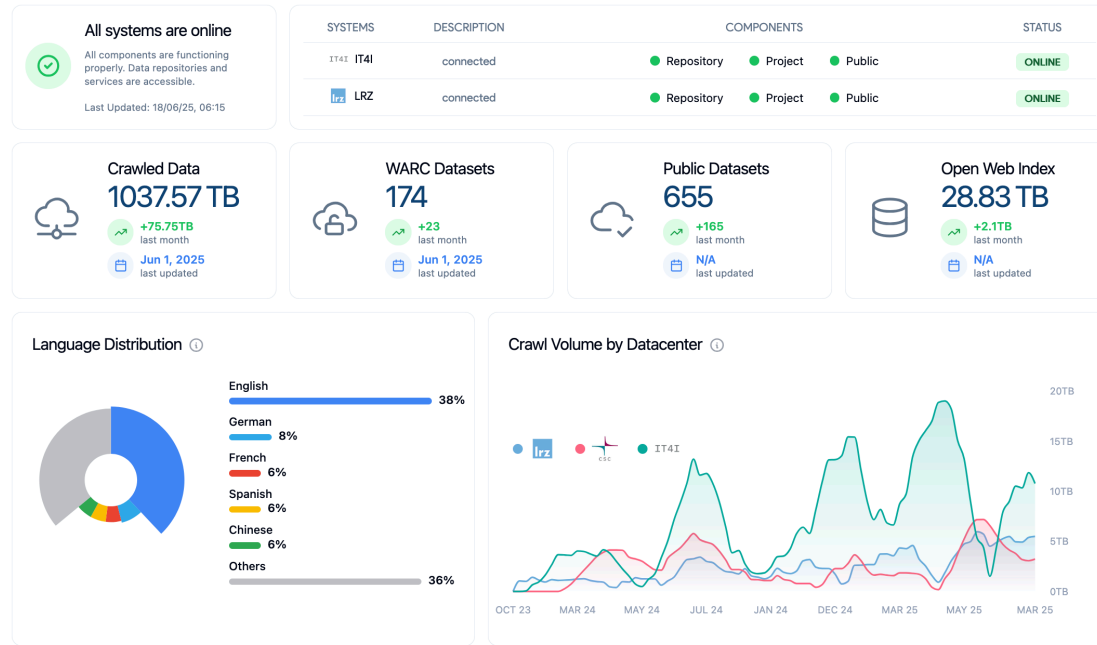


# IN DETAIL

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# Accessing the Open Web Index

# Dashboard — Statistics & Datasets in Detail



## OWI Statistics Tab

- Repository health per data center (IT4I · LRZ)
- Key indicators: pipeline runs, HPC core-hours consumed, avg. duration, active pipelines
- Crawled data (1.3 PB), WARC datasets, public datasets, index size
- Language distribution chart + crawl volume timeline per DC
- Dataset collection distribution by resource type (main / GPU / curlie / embeddings)

## OWI Datasets Tab

- Filter: datacenter · format (WARC / Parquet / CIFF / Embeddings) · language · date range
- Infinite-scroll dataset cards: title, date, DC, public/private badge, file listing
- Parquet file preview in-browser; download via LEXIS Portal or owilix remote pull

# Dashboard — Corpora, Models & Search

## OWI Corpora

Curated, administrator-prepared collections for distribution:

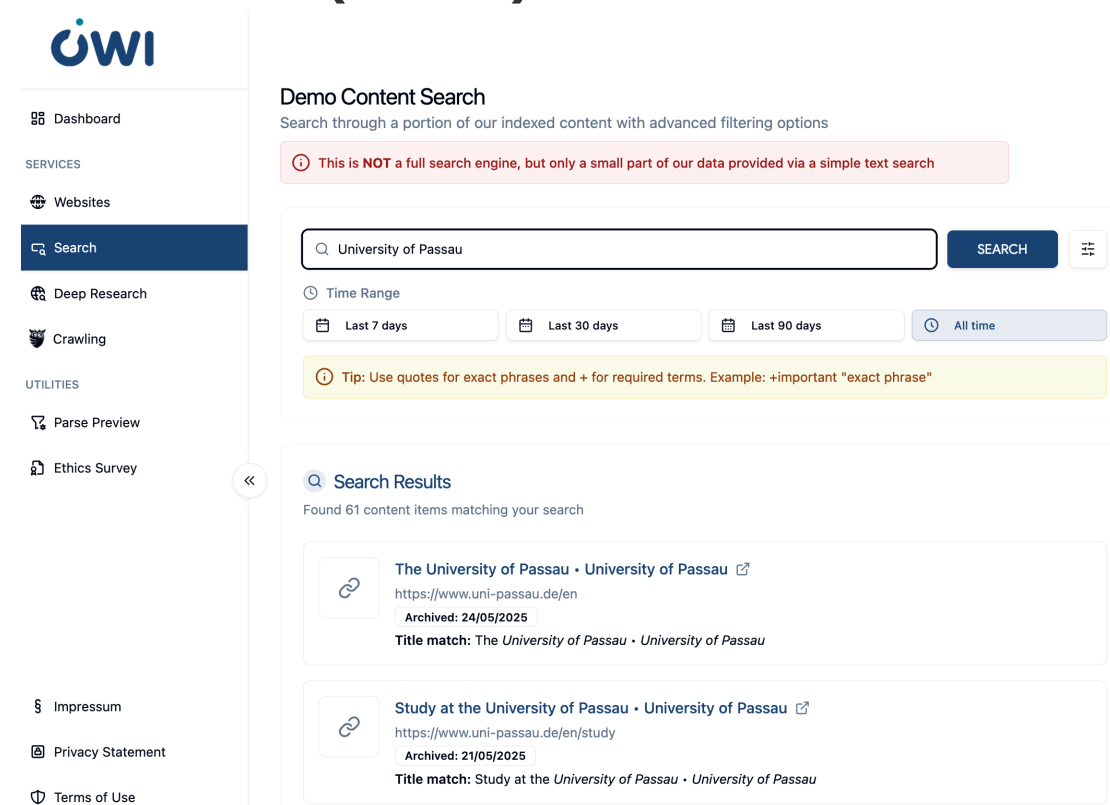
- Licensing metadata per corpus (creator, terms, full licence text)
- LEXIS integration for storage, access control and metadata
- Includes domain-specific and project-specific collections

## OWI Models

Registry of language models for web search:

- Browser-runnable models via WebAssembly (Wllama library)
- Cloud-based models (OpenAI, DeepSeek) for heavy tasks
- Model cards with training data, benchmarks,  and GGUF file listing

## OUR Search (Demo)



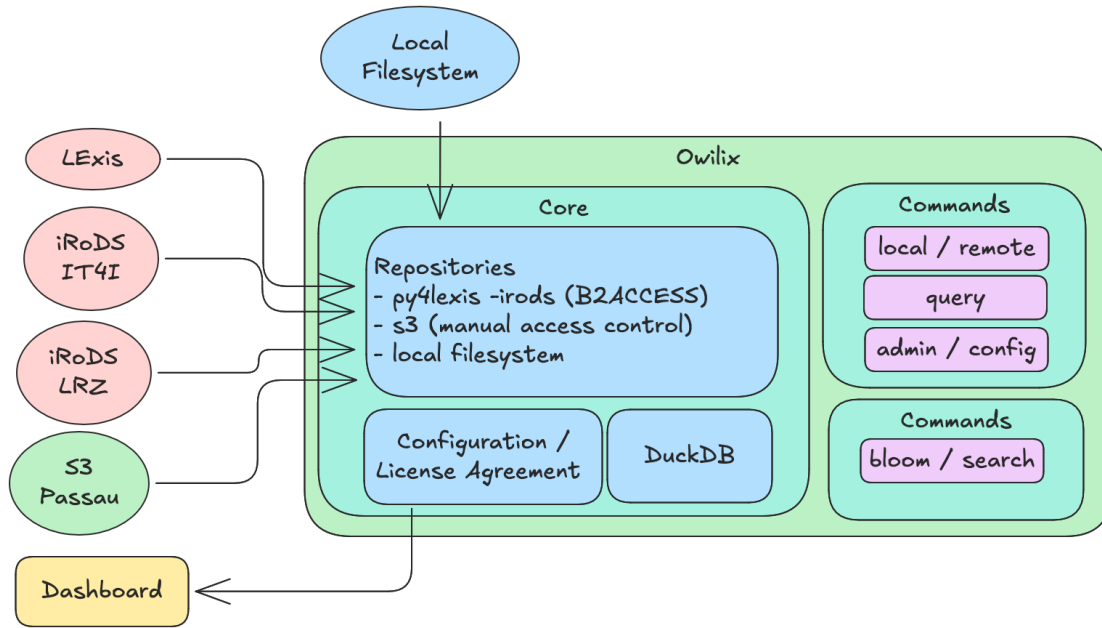
The screenshot shows the OWI Search Demo interface. On the left is a sidebar with the OWI logo and a navigation menu: Dashboard, SERVICES (Websites, Search, Deep Research, Crawling), UTILITIES (Parse Preview, Ethics Survey), Impressum, Privacy Statement, and Terms of Use. The 'Search' option is highlighted. The main content area is titled 'Demo Content Search' and includes a subtitle 'Search through a portion of our indexed content with advanced filtering options'. A red warning box states: 'This is NOT a full search engine, but only a small part of our data provided via a simple text search'. Below this is a search bar with the text 'University of Passau' and a 'SEARCH' button. A 'Time Range' filter is set to 'All time', with other options being 'Last 7 days', 'Last 30 days', and 'Last 90 days'. A yellow tip box says: 'Tip: Use quotes for exact phrases and + for required terms. Example: +important "exact phrase"'. The 'Search Results' section shows 'Found 61 content items matching your search'. Two results are visible: 'The University of Passau · University of Passau' with URL 'https://www.uni-passau.de/en' and 'Archived: 24/05/2025', and 'Study at the University of Passau · University of Passau' with URL 'https://www.uni-passau.de/en/study' and 'Archived: 21/05/2025'. Both results show a 'Title match'.

Simple + advanced field-specific search over a subset of the index; time-range filtering; result cards with thumbnail, title, URL

## Deep Research (Experimental)

AI-powered multi-step research: configurable depth + breadth, local or cloud LLM, interactive clarification mode

# owilix — The Open Web Index Client



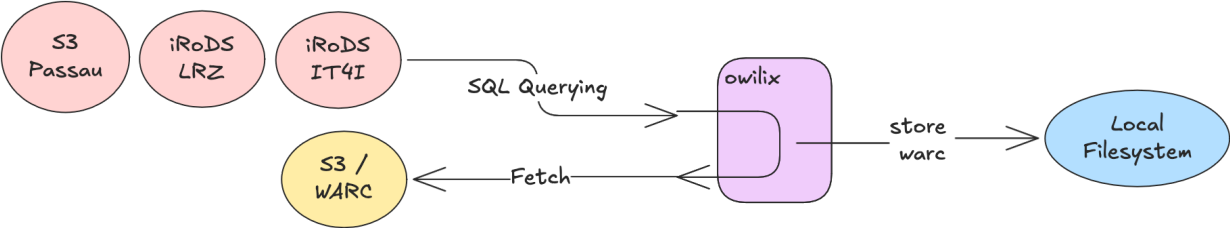
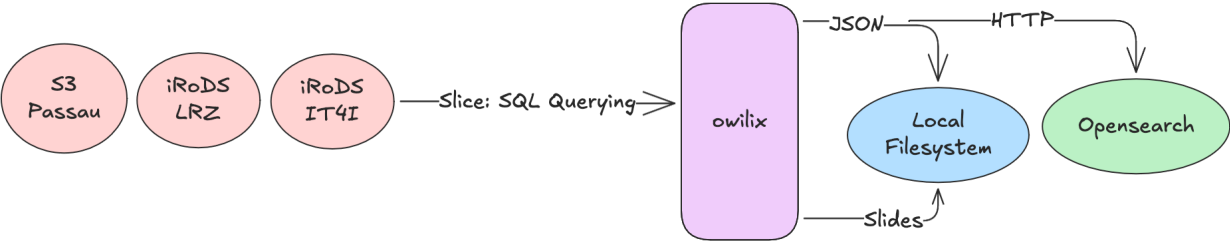
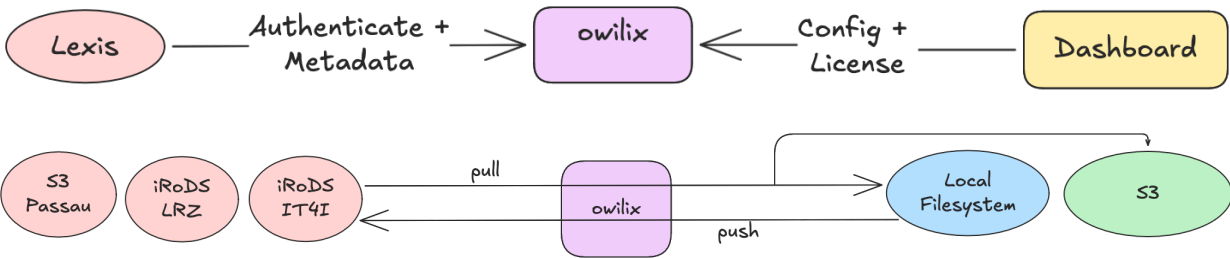
Four main workflows:

1. **Authentication** — B2ACCESS via LEXIS + Dashboard config (OpenID Connect / Keycloak)
2. **Sync Datasets** — pull/push between federated iRODS/S3 storage and local filesystem
3. **Query / Slice** — client-side SQL queries with JSON, HTTP, OpenSearch output
4. **WARC Access** — fetch original WARC files via SQL querying

Slice the index by:

- **Domain** (.de, .at, .fr, ...)
- **Topic** (Curlie.org hierarchy labels)
- **Language** (filename-based selection)
- **Sitelist** (CSV of URLs / domains / TLDs)
- **Structured data** / microdata type
- **Outlinks** for link graph analysis

# owilix Workflows



# Remote Index — Query Without Downloading

## What is the remote index?

A DuckDB / Apache Spark-queryable representation of the OWI — Parquet partitions aligned with CIFF shards, accessible from remote clients without downloading the full index.

## Example queries

```
-- Count documents per language
SELECT language, COUNT(*) as docs
FROM read_parquet('remote://lrz:latest/main/**')
GROUP BY language ORDER BY docs DESC;

-- Filter by domain
SELECT url, title FROM ... WHERE domain = 'wikipedia.org';
```

## Use cases

- Analytics and statistics on the live index
- Lightweight RAG pipeline ingestion
- Ad-hoc exploration before committing to a full  shard download

## OUR<sup>2</sup>S — Online Search API

OUR<sup>2</sup>S (built on Vespa) extends the remote index with a full REST search API:

- Structured document model: full page · snippets · embeddings
- Keyword, semantic and hybrid ranking modes
- Multi-index federation across data centers
- No local index needed — query directly over network

→ OUR<sup>2</sup>S details in the dedicated OUR<sup>2</sup>S column

## Access summary

| Mode   | Tool      | Authentication       |
|--------|-----------|----------------------|
| Browse | Dashboard | Anonymous / B2ACCESS |

| Mode                  | Tool                   | Authentication      |
|-----------------------|------------------------|---------------------|
| Download              | owilix                 | B2ACCESS<br>(LEXIS) |
| SQL queries           | DuckDB<br>remote       | B2ACCESS            |
| REST search           | OUR <sup>2</sup> S API | API key             |
| Conversational<br>RAG | MosaicRAG              | Open /<br>B2ACCESS  |

# MosaicRAG — Building RAG Systems on OWI

## From OWI Datasets to Conversational Search

MosaicRAG consumes OWI daily datasets directly — CIFF inverted index files, Parquet metadata records, and dense embeddings (published since Dec 2025) — and exposes them through a modular conversational interface.

### Pipeline for building your own MosaicRAG instance:

1. Download OWI slice with `owilix` (topic, domain, or language filter)
2. Ingest CIFF → inverted index; Parquet → DuckDB; embeddings → ChromaDB
3. Configure retrieval pipeline: sparse, dense, or hybrid
4. Choose reranker (TF-IDF · Embedding cross-encoder · Tournament LLM · Group LLM)
5. Optionally add summarization and relevance marking

Full documentation: [openwebindex.eu/book](https://openwebindex.eu/book) — "Building vertical search engines with MOSAIC"

### Supported LLMs

| Task              | Models                    |
|-------------------|---------------------------|
| Reranking         | gemma2, qwen2.5, llama3.1 |
| Summarization     | + DeepSeekv3              |
| Relevance marking | gemma2, qwen2.5, llama3.1 |

### Live Demo

Try conversational search on the Open Web Index at [mosaicrag.ows.eu](https://mosaicrag.ows.eu)

# Using the Open Web Index — Applications

# Using the Open Web Index — Applications

## Realised Applications

- **Mosaic & MosaicRAG** — modular vertical search framework; science, health, arts demos → [mosaic.ows.eu](https://mosaic.ows.eu)
- **Earth Observation Search** — RAG QA + geo-contextualised search (DLR / Radboud)
- **Multilingual Scientific Index** — enriching Finland's Research.fi portal (CSC)
- **Restaurant Recommender** — privacy-preserving mobile app; 214-user pilot (A1 Slovenia)
- **Argumentation Search** — args.me extended with OWI pro/contra index →

## Community Applications (FSTPs)

- **AKSE** — argumentation knowledge graphs
- **VERITAS / DEXAI** — verification-oriented RAG
- **OMMS** — enriching OpenStreetMap with POI metadata
- **DTCommerce** — SME e-commerce product enrichment
- **CIFFIL** — cross-collection index for municipal search
- **TILDE** — trustworthy health domain search

The OWI enables a full ecosystem of vertical search engines, RAG systems,

[args.me/owi](https://args.me/owi)

- **CERN Search** — institutional search + disability knowledge search (Nooon)

and domain-specific applications

# IN DETAIL

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# MosaicRAG — Conversational Search on the OWI

**Modular RAG framework** extending MOSAIC with LLM-based capabilities and dense retrieval (WEBIS / TU Graz)

- **Two retrieval paths:** inverted index (CIFF) + dense retrieval
- **4 reranking strategies:** TF-IDF, Embedding cross-encoder, Tournament-style (LLM pairwise), Group-style (LLM window)
- **LLM backends:** self-hosted on-premise models for summarization
- **Conversational search:** follow-up questions about retrieval results; dual-window UI (search results + chat)
- **Workflow:** define retrieval pipeline → issue query → converse about results

## Key Features

- Relevance marking — LLM highlights relevant passages within documents
- Summarization — user-provided instructions applied per document
- Domain-specific indices: science, health, arts (8M+ docs)
- Fully open-source modular pipeline

## Data Pipeline

OWI daily datasets → CIFF (inverted index) + Parquet (metadata) + dense embeddings (since Dec 2025) → MosaicRAG

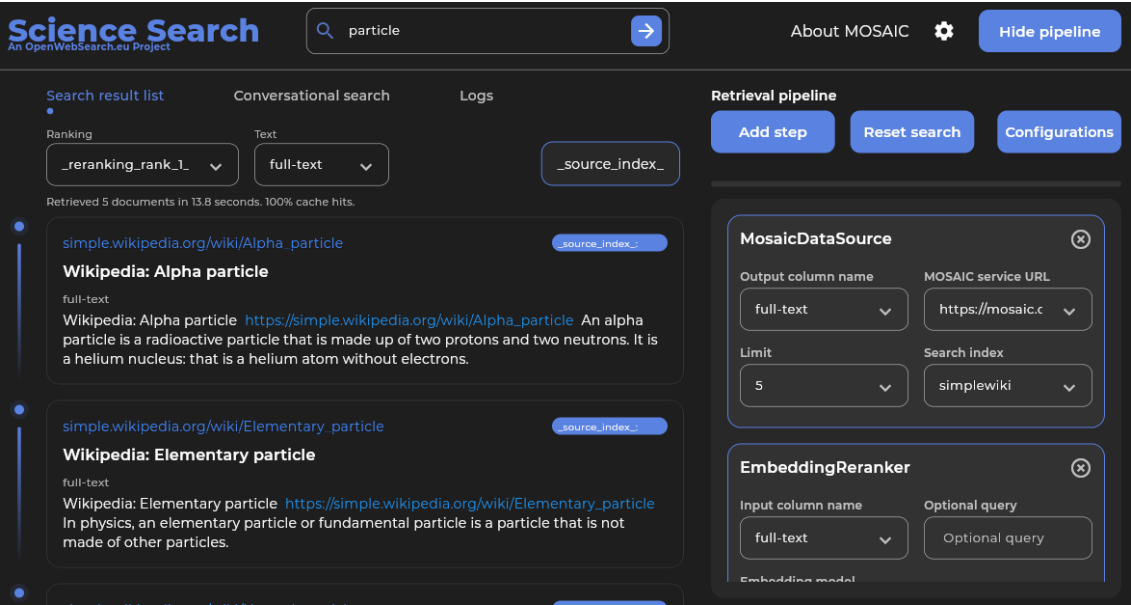




# Mosaic Search Applications · Earth Observation Search

## Mosaic Vertical Search (TU Graz)

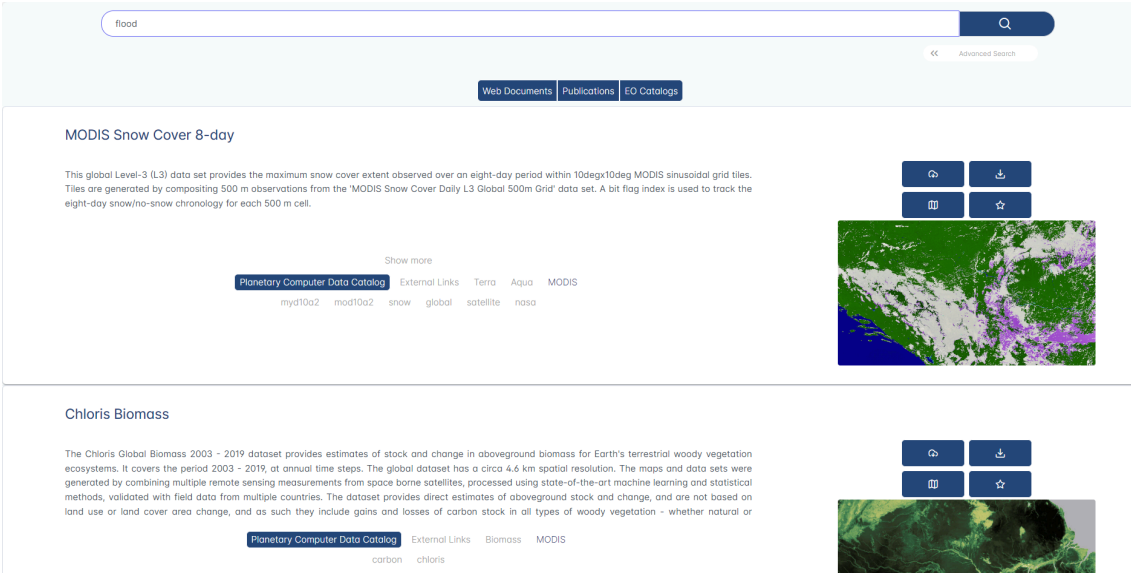
Modular framework for building vertical search engines on OWI slices — domain-specific indices for science, health, arts (8M+ docs)



Demo: [mosaic.ows.eu](https://mosaic.ows.eu) · [mosaic.ows.eu/demos](https://mosaic.ows.eu/demos)

## Earth Observation Search (DLR / Radboud)

RAG-based QA over scientific papers, datasets & web content; geo-contextualized multi-genre search on interactive maps; KG with 2M+ nodes (OpenAlex, PANGAEA, DLR Geoservice)



Demo: [eo-rag-demo.2.rahtiapp.fi](https://eo-rag-demo.2.rahtiapp.fi) · [ows-demo.2.rahtiapp.fi](https://ows-demo.2.rahtiapp.fi)

# Multilingual Scientific Index · Restaurant Recommender

# Multilingual Scientific Index (CSC)

Enriches Finland's Research.fi portal with OWI web data — Finnish & English science content classified by type (publication, blog, event); embedded directly in national research portal

| Energy-centric optimization of large-scale additive manufacturing deployment for sustainability through electrification |                                                                                                                                                                                                                                                                                                                               |                                                         |           |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-----------|
| Rahoitetun hankkeen kuvaus                                                                                              | The employment of additive manufacturing (AM), commonly called 3D printing, in the industrial sector has been growing for many years. As the technology matures, it is expected that its use will become widespread and substitute other established forms of production. This project is...<br><a href="#">Näytä enemmän</a> |                                                         |           |
| Aloitusvuosi                                                                                                            | 2026                                                                                                                                                                                                                                                                                                                          |                                                         |           |
| Päättymisvuosi                                                                                                          | 2029                                                                                                                                                                                                                                                                                                                          |                                                         |           |
| Myönnetty rahoitus                                                                                                      | Humberto Almeida Junior                                                                                                                                                                                                                                                                                                       | Lappeenrannan–Lahden teknillinen yliopisto LUT          | 305 350 € |
| Rooli Suomen Akatemian konsortiossa                                                                                     | Partneri                                                                                                                                                                                                                                                                                                                      |                                                         |           |
| Muut osapuolet                                                                                                          | Partneri                                                                                                                                                                                                                                                                                                                      | Aalto-yliopisto (372725)                                | 298 731 € |
|                                                                                                                         | Johtaja                                                                                                                                                                                                                                                                                                                       | Lappeenrannan–Lahden teknillinen yliopisto LUT (372723) | 395 919 € |
| Rahoittaja                                                                                                              | Suomen Akatemia                                                                                                                                                                                                                                                                                                               |                                                         |           |
| Rahoitusmuoto                                                                                                           | Suunnattu akatemiahanke                                                                                                                                                                                                                                                                                                       |                                                         |           |
| Haku                                                                                                                    | <a href="#">2025 Tulevaisuuden kestävien energiaratkaisujen tutkimuksen haku 2025</a>                                                                                                                                                                                                                                         |                                                         |           |
| Muut tiedot                                                                                                             |                                                                                                                                                                                                                                                                                                                               |                                                         |           |
| Rahoituspäätöksen numero                                                                                                | 372724                                                                                                                                                                                                                                                                                                                        |                                                         |           |
| Tieteenalat                                                                                                             | Kone- ja valmistustekniikka                                                                                                                                                                                                                                                                                                   |                                                         |           |
| Tutkimusalat                                                                                                            | Kone- ja valmistustekniikka                                                                                                                                                                                                                                                                                                   |                                                         |           |

### Myöntöön liittyvä verkkosivu 1

Rahoitusmyöntöön ei ole tarjolla linkkejä tässä portaaliissa

### Viittaukset muissa palveluissa

Viittauksia yhteensä: 20

| Kategoria | Prosentti | Määrä |
|-----------|-----------|-------|
| Other     | 40%       | (8)   |
| Article   | 30%       | (6)   |
| News      | 20%       | (4)   |
| Blog      | 10%       | (2)   |
| Event     | 0%        | (0)   |

### Hakutulokset 1 - 5

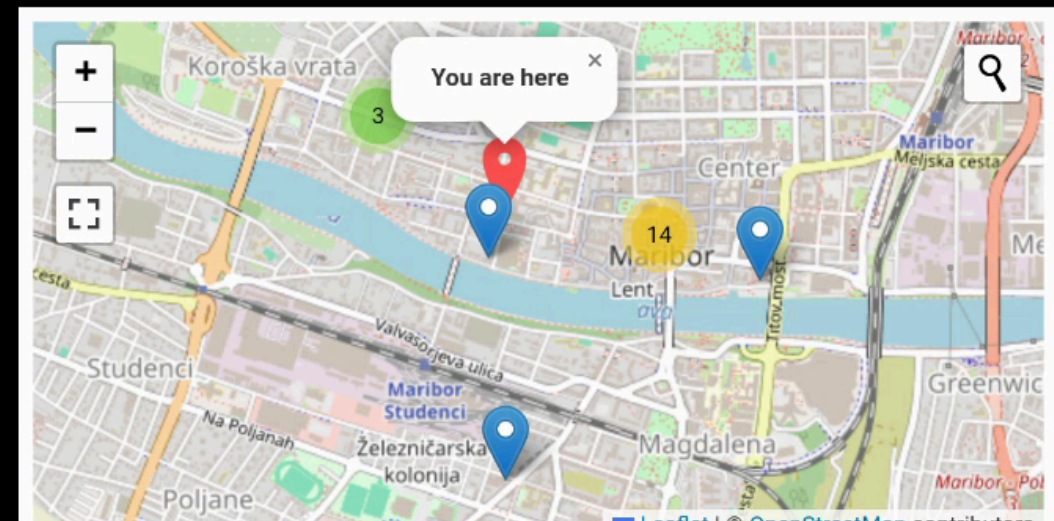
- <https://www.industryweek.com/...>  
3 Ways Additive Is Shaping the Factory of the Future | IndustryWeek
- <https://www.industryweek.com/...>  
These Industries are the Future of Additive Manufacturing | IndustryWeek
- [https://global-sci.com/...](https://global-sci.com/)  
Global Science Press: A Review of Process Optimization for Additive Manufacturing Based on Machine Learning
- [https://www.engineering.com/...](https://www.engineering.com/)  
Customized Production Will Become Scalable with 3D Printing - Engineering.com
- [https://axial.acs.org/...](https://axial.acs.org/)  
Welcome the Newest Associate Editors of ACS Publications Journals: Q3 2018 | ACS Publications Chemistry Blog

< 1 2 3 4 >

## Restaurant Recommender (A1 Slovenia)

Privacy-preserving mobile app; on-device personalization via query imputation; LLM-extracted restaurant features from OWI; 5-month pilot with 214 users — overall satisfaction 4.2 / 5

# Restaurant Recommender



Or search for restaurants in 1 km radius

Select City ▼

Bascarsija grill food



## Baščaršija

[Visit Restaurant Page](#)

Poshtna ulica 8, Maribor 2000 Slovenia 📞 +386 2 250 63 59

Cuisines: Barbecue; European; Grill; Slovenian; Eastern European Meals: Lunch; Dinner

Price: \$4 - \$13 Features: outdoor seating;

## Okrepcevalnica Sarajevo

[Visit Restaurant Page](#)

Gospodsvetska cesta 43c, Maribor 2000 Slovenia 📞 +386 2 252 34 31

Cuisines: Barbecue; European Meals: Lunch; Dinner Price: \$2 - \$11 Features: outdoor seating;

# Argumentation Search · CERN Search Applications

## Argumentation Search (WEBIS)

args.me extended with OWI: pro & contra arguments on 31 controversial topics; argument quality scoring, key point matching, temporal trend analysis

## CERN Search Applications

- **Institutional**: 25K+ pages crawled from CERN internal web; full OWS pipeline → MOSAIC index for AccGPT knowledge base
- **Nooon**: disability knowledge search (2.59M docs) for HR & Diversity & Inclusion





We should abolish the right to keep and bear arms

violence

All

Search

520 results

46 ms

Pro-Arguments

Con-Arguments

ranking score: 9.051

source

Argumentative Segment

Since the start of the year, there have been 80 mass shootings and more than 3,300 deaths due to gun **violence** nationwide, according to the Gun **Violence** Archive. We don't need any more thoughts and prayers from elected leaders. We shouldn't have any more moments of **silence** or vigils. We need action.

Key Point

Quality Score: 0.880

Gun ownership allows for mass-shootings/general gun violence

ranking score: 8.636

source

Argumentative Segment

"Any successful strategy to reduce gun **violence** requires preventing the diversion of lawful firearms into unlawful commerce," Steven Dettelbach said at the time. "Once there, these firearms end up in the hands of people who are sometimes **violent** criminals and intend to do harm to the people with whom we live, the innocent people who are victims and survivors of gun **violence**."

Key Point

Quality Score: -0.690

Guns can fall into the wrong hands

ranking score: 11.986

source

Argumentative Segment

Councilman Johnson, repeat after me: **Violence** is **Violence**. The number one cause of **violence** is **violent** people. While semantically there is an argument that some people aren't **violent** until provoked, drunk and/or stoned, exposed to constant **violence** in entertainment media, video games etc., the fact remains that **violence** always, 100% of the time, involves at least one **violent** person. Blaming guns for **violence** is like blaming hammers for shoddy home construction.

Key Point

Quality Score: 0.660

ranking score: 8.489

source

Argumentative Segment

No matter how many times actual, real-life case studies are laid out for Cleveland's elected officials, they refuse to recognize that the only PROVEN method of reducing **violence** is removing **violent** persons from circulation, either through imprisonment or by vibrantly empowering the potential victims to remove the **violent** person when confronted by one.

Key Point

Quality Score: 0.870

Gun ownership promotes self protection

Demo: [args.me/owi](https://args.me/owi) · [args.me](https://args.me)

MOSAIC

ows-lb-2.cern.ch/ui/

80%

☆

⌵

MOSAIC Search

Search term: disability

Search

Index Info

Geo Filter:

West:

East:

North:

South:

Index:

☐ default / all

☒ disability-index

Language:

☒ default / all

☐ English

☐ German

Limit:

☒ default / 20

☐ 10 items

☐ 50 items

☐ 1,000,000

Keyword:

Search URL: https://ows-lb-2.cern.ch/mosaic/search?q=disability&index=disability-index

Search result for term: "disability"

Index: disability-index

Number of items: 20

Cockrell Hill Social Security Disability Attorneys | OneCLE Texas Attorney Directory

Kraft Dallas, TX Social Security Disability Lawyer with 54 years of experience (214) 999-9999 2777 N Stemmons Fwy Suite 1300 Dallas, TX 75207 Free ConsultationSocial Security Disability and Personal Injury Baylor Law School Show Preview View Website

Metadata: language:eng, word count:4273, index date:2025-08-10 00:13

Locations:

Keywords: Attorney • Attorneys • Social Security Disability • Cockrell Hill • Texas • <https://lawyers.onecle.com/lawyers/social-security-disability/texas/cockrell-hill>

Insurance Coordinator Manual

Partial disability must result from the same condition as the total disability. Proof of partial disability must be submitted within 31 days of the date the employee's total disability period ends. Pa

Metadata: language:eng, word count:20858, index date:2025-08-10 00:51

Locations:

Keywords: <https://oklahoma.gov/egid/insurance-coordinators/manual.html>

House of Lords/Commons - Joint Committee on the Draft Disability Discrimination Bill - First Report

Disability Rights Commission, DDB 1, para 10.6.1. British Council of Disabled People, DDB 6, para 9. Law Society, DDB 15, p7. 15(a): The Government agrees with the recommendations of the DRTF that MPs

Metadata: language:eng, word count:12308, index date:2025-08-10 00:25

Locations:

Keywords: <https://publications.parliament.uk/pa/jt200304/jtselect/jtdisab/82/8222.htm>

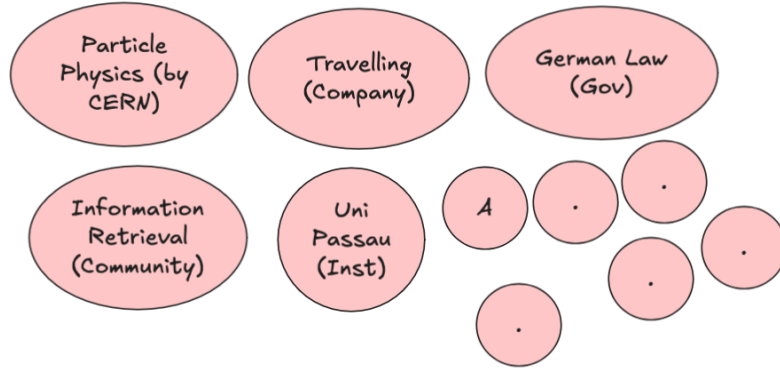
UKRI diversity data for funding applicants and awardees 2020 to 21 update – UKRI

After removing this opportunity, the award rate for fellows reporting a disability was 22% compared with

## Third-Party Projects (FSTPs)

- **AKSE** — argumentation knowledge graph for structured deliberation
- **VERITAS / DEXAI** — verification-oriented RAG for sensitive geopolitical contexts (Ukraine)
- **OMMS** — enriching OpenStreetMap POI with OWI structured data
- **DTCommerce** — product extraction & enrichment pipeline for SME e-commerce
- **CIFFIL** — cross-collection index interoperability for municipal search (Dutch pilots)
- **TILDE** — trustworthy health search with bias detection and entity-level retrieval

# Towards Federated RAG Systems



- RAG / Agentic systems require focused, high-quality collections
- Web information redundancy increases due to LLMs — no one-size-fits-all retrieval
- OpenWebSearch.eu supports low-cost creation of topic-specific RAG systems

## Key ideas:

- Retrieve from well-designed RAG systems instead of generic search engines
- Topic-specific, deduplicated and curated content
- Domain-optimized ranking, aggregation and summarization
- Lightweight browser-based federation with personal data integration

**OwlerLite** (WWW 2026) — privacy-respecting browser extension for scope- and freshness-aware retrieval: user-defined **scopes** (trusted sources), semantic freshness detection via SimHash + vector similarity, Scope Fidelity improved from 0.64 → **0.83** (negligible NDCG loss)

# Using the Open Web Index — Data & Analytics

# Using the Open Web Index — Data & Analytics

## Published Datasets

| Dataset         | Scale                 | Use case                        |
|-----------------|-----------------------|---------------------------------|
| WebFAQ          | 96M Q&A, 75 langs     | QA training, cross-lingual IR   |
| WebFAQ 2.0      | 198M pairs, 108 langs | Dense retrieval, hard negatives |
| Imprints        | 5.25M hosts, geocoded | Enterprise analytics, Eurostat  |
| German Commons  | 154B tokens           | German LLM pre-training         |
| OWS-Curlie-2025 | 20.7M docs + qrels    | IR evaluation benchmark         |

All datasets available at [openwebindex.eu/corpora](https://openwebindex.eu/corpora)

## Analytics Potential

- **Official Statistics:** enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring:** supply chain analysis, news monitoring (JRC)
- **AI Training Data:** domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks:** web-scale IR test collections with relevance assessments
- **Trend Analysis:** temporal web monitoring at Petabyte scale

The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

# IN DETAIL

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# WebFAQ — Multilingual FAQ from the Web

- **96 million** Q&A pairs across **75+ languages**
- Extracted from structured FAQ markup (JSON-LD, Microdata) in the Open Web Index
- Fields: question, answer, URL, origin, topic, question type
- Parquet format, hosted on Hugging Face

**Use cases:** multilingual QA training, information retrieval, cross-lingual transfer learning

[huggingface.co/datasets/PaDaS-Lab/webfaq](https://huggingface.co/datasets/PaDaS-Lab/webfaq)

| Top languages | Pairs     |
|---------------|-----------|
| English       | ~30M      |
| German        | ~8M       |
| French        | ~6M       |
| Spanish       | ~5M       |
| ...           | 75+ langs |



# WebFAQ 2.0 — Hard Negatives for Dense Retrieval

- **198 million** FAQ-based Q&A pairs across **108 languages**
- **14.3 million** bilingual aligned QA pairs — largest FAQ-based resource
- **1.25 million** queries with mined hard negatives (20 languages)
- 200 cross-encoder scored negatives per query

## Key contributions:

1. Novel web crawling for diverse content extraction
2. Hard negatives for dense retriever training
3. Contrastive Learning & Knowledge Distillation
4. Open datasets and training scripts

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*Dinzinger, Caspari, Salman, Topi, Mitrović & Granitzer. "WebFAQ 2.0: A Multilingual QA Dataset with Mined Hard Negatives for Dense Retrieval." arXiv:2602.17327, 2025.*

# Imprints Dataset — Geospatially Grounded Web Curation

- **5.25 million** hosts, **47.8 GB**
- Each website annotated along two dimensions:
  - **Topical classification** via LLM (Foursquare category hierarchy)
  - **Geographic grounding** from legal imprints via geocoding
- **3.08 million** websites geocoded from extracted addresses
- Page types: main page, imprint, contact, about us, privacy, terms

| Country     | Geocoded  | Share |
|-------------|-----------|-------|
| Germany     | 2,636,913 | 85.5% |
| Austria     | 223,805   | 7.3%  |
| Switzerland | 160,683   | 5.2%  |

| Top-Level Topic         | Share |
|-------------------------|-------|
| Business & Professional | 41.6% |
| Community & Government  | 16.2% |
| Arts & Entertainment    | 12.2% |

# German Commons — 154B Tokens for German LLMs

- 154.56 billion tokens, 35.78 million documents
- Aggregated from 41 diverse sources under open licenses
- Quality-filtered, deduplicated, OCR scored

| Domain             | Tokens |
|--------------------|--------|
| News Commons       | 72.67B |
| Cultural Commons   | 54.49B |
| Web Commons        | 19.89B |
| Political Commons  | 3.57B  |
| Legal Commons      | 2.99B  |
| Scientific Commons | 0.84B  |
| Economics Commons  | 0.11B  |

Gienapp, Schröder, Schweter et al. "The German Commons — 154 Billion Tokens of Openly Licensed Text for German Language Models." arXiv:2510.13996, 2025.



# OWS-Curlie-2025 — Web Search Test Collection

- **20.7 million** documents from 6 months of OWI crawl data (March–August 2025)
- English documents from the Curlie subcollection (websites labeled in the Curlie directory)
- Three evaluation splits with student-provided topics and relevance assessments
- Jina V3 embeddings, ClFF indexes for sparse retrieval
- Accessible via owilix CLI and ir-datasets-owi Python package

| Split       | Documents  | Embeddings | Qrels |
|-------------|------------|------------|-------|
| all         | 20,746,181 | --         | --    |
| radboud-val | 63,639     | yes        | yes   |
| radboud     | 90,852     | yes        | --    |
| jena-kassel | 77,382     | yes        | --    |

[openwebindex.eu/corpora](https://openwebindex.eu/corpora) · License: OWIL v1.0



# Using the Open Web Index — Ideas

# Potential Application Domains

Some application scenarios discussed at the start of OpenWebSearch.eu are now in reach:

## Web-Analytics for Official Statistics

**Eurostat:** enterprise information extraction from legal imprints — who is doing business where?



**Destatis:** statistical data extraction from web sources to complement national statistics



## Media Monitoring & Intelligence

**JRC Media Monitor:** monitoring media ecosystems and supply chain signals at European scale (*in start phase*)



## A Live Use Case: EU Toy Safety & the Open Web Index

Can the OWI be used for EU product market surveillance? In preparation of the talk I ran a live exploratory analysis using the OUR<sup>2</sup>S structured-data API plus some AI Agents.

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*"Make an analysis of the OURS API for the OWI corpus: find children's toys and assess how they appear in the dataset — particularly those potentially violating EU regulations."*

Prompt used in JetBrains's Junie AI System with Gemini 3.1 Flash

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*"Make an analysis of the OURS API for the OWI corpus: find children's toys and assess how they appear in the dataset — particularly those potentially violating EU regulations."*

Prompt used in Jetbrain's Junie AI System with Gemini 3.1 Flash

## The corpus

| Schema              | Documents |
|---------------------|-----------|
| Web pages           | 310 M     |
| Structured entities | 137.5 M   |
| Unique hosts        | 10 M      |

## Entities Found

- **27 M Products** — largest single type
- 26 M Organisations
- 18 M Articles
- 3.8 M Local Businesses
- 2.8 M Events

→ Products are the single largest entity type — directly usable for market analysis.

# A Live Use Case: Products at Scale — Toys in the Index

## Pricing & geography

- ~15 M entities carry price data
- **3 M EUR-priced** products → significant EU market coverage
- Also: USD 2.2 M · UAH 1.9 M · BRL 1.2 M

## Toy query results (live)

- "children toy" → **102,530 products**
- With pricing → **51,384** (50 % have structured prices)
- "Spielzeug Kinder" → **7,888** German results

## Sample items found

- *"1:24 die-cast metal sports car"* — \$90.99
- *"Kids Microscope 1000× Magnifier"* — \$59.00
- *"48-teiliges Einhorn Prinzessin Set"* — €8.44

## Structured fields available per product

- Name, brand, GTIN/EAN
- Price, currency, availability
- Description, images, ratings

~50 % of toy products include structured price data, reflecting real e-commerce markup adoption.

~92 % of all entities **lack location data** — a significant data-quality finding.

Whole analysis took 20 minutes in parallel to working on this presentation.

## A Live Use Case: Conclusion

### What the OWI provides today

- Structured Microdata with some features
  - GTIN/EAN presence assessment
  - Brand & manufacturer data
- Completeness and parsing ok, but should be improved

### What would be missing for full EU compliance

- CE marking (*not in schema.org*)
- Age suitability / safety warnings
- Material composition
- ...

Could be added to parser

The OWI is a **foundation for such a use case** — Without an aggregated web index, such an analysis would not be possible. Of course you can use commercial search engines for such tasks, but they are not transparent and you don't know what data they use.

# --

## Other Research Directions

### Next-Generation Retrieval

- Federated RAG across distributed OWI slices and 3rd-party corpora
- Agentic search orchestration for autonomous knowledge discovery
- Multimodal indexing: images, tables, charts alongside text
- Temporal web search: tracking how the web changes over time

### AI & LLM Infrastructure

- Domain-specific corpora for sovereign European LLM training
- Retrieval-Augmented Generation benchmarks on OWI data
- Open evaluation infrastructure for IR and NLP research
- AI Agents & Search - actively pursued by Uni Passau
- **Multi-loop LLM orchestration:** classify incoming tasks, represent task types as vectors, and use cross-encoder-style scoring to route or sequence specialised LLM loops for the identified tasks
- **Stacked Kohonen embeddings:** map each sentence token to the top-n matching SOM codebook vectors, use their grid positions as explicit embedding coordinates, and condition the next token's SOM lookup on the previous token's match to form a joint sequence vector
- **Capability allocation across parameter space:** which capabilities — knowledge, reasoning, behaviour — should live in *model parameters* vs. *indexed memory* vs. *interaction traces*? Open design question; envisioned



figure maps capability types across the parameter / retrieval / trace space (parked from the TPC26 "Open System Question" slide)

## Societal & Policy Applications

- **Information quality monitoring:** detecting coordinated inauthentic behaviour, SEO spam
- **Digital literacy tooling:** transparent, inspectable search for education
- **Regulatory compliance:** evidence-based auditing of search engine behaviour
- **Digital Sovereignty:** European alternative to US-dominated search infrastructure

The OWI is an open infrastructure — we invite researchers, businesses, and public institutions to build on it

We are looking to be part of interesting research projects to advance the OWI.

We hope to build an open ecosystem that puts open, fair and unbiased information exchange and societal values before commercial interest.

# Ethical, Legal & Societal Aspects (ELSA)

# ELSA Framework — Values, Risks & Mitigation

## 10 Ethical Values

1. Transparency
2. Safety & Security
3. Openness
4. Responsibility
5. Privacy
6. Justice
7. Trustworthiness
8. Accountability
9. Sustainability
10. Sovereignty / Autonomy

## 5 Ethical Risks

1. Control of indexed websites & take-down requests — censorship vs. coverage
2. Flawed content in the OWI — false, misleading, offensive material
3. Inaccurate metadata — errors in ingestion or preprocessing
4. Misuse by third parties — privacy violations, unethical applications
5. Unsuitable governance — undemocratic control, security breaches

## ELSA Deliverables

Three iterations (D6.1 → D6.2 → D6.3), final version subsumes all. Developed with:

- 3 multi-day ethics workshops (CAIS / Stiftung Mercator)
- 3 legal FSTP projects (LISA, LOREN, ALMASTIC)
- OSF Working Groups on Ethics & Law

## 6 Stakeholder Groups

End users · Index users · Developers · Researchers & institutions · Content providers · Governance

# Legal Challenges & Regulatory Landscape

## Regulatory Framework

- **8+ EU regulations** apply: GDPR, Copyright Directive, DSM Directive, DSA, DMA, Data Act, Data Governance Act, AI Act
- **Copyright is the biggest challenge** — impacts crawling, redistribution, and LLM training in Europe
- No guiding court decisions yet — first time an open web index operates in the EU
- DSA classification unclear — OWI does not fit neatly as hosting service, platform, or search engine
- **AI Act**: OWI classified as "**limited risk**" — research exemption applies during project; post-project needs reassessment

## Legal FSTP Projects (published on Zenodo)

- **LISA** — Legal Framework for OWI (Matthias Wendland)
- **LOREN** — Legal Open European Web Index (Paul Johannes, Maxi Nebel)
- **ALMASTIC** — Assessing Legal Risks (Kai Erenli, Andreas Gruber)

## Key Legal Risks

- Personal data in crawled content (GDPR Art. 6(1)(f) legitimate interest)
- Copyright infringement — images, paywalled articles, plain text
- Liability as digital intermediary
- Illegal content obligations
- LLM training restrictions (no\_genai flag)

# 10 Technical & Organisational Mitigation Measures

| #   | Measure                         | Key Mechanism                                                        |
|-----|---------------------------------|----------------------------------------------------------------------|
| M1  | Website Management              | robots.txt opt-out, blacklisting, quality prioritisation             |
| M2  | Take-down Management            | Dashboard submission, logged removal, blacklist propagation          |
| M3  | Content Analysis                | Quality scoring, trigger warnings, ad detection, plugin architecture |
| M4  | Access Control                  | B2ACCESS registration, immediate blocking of misuse                  |
| M5  | Transparency Guideline          | 27-aspect guideline for third-party search apps                      |
| M6  | Data Protection & Privacy       | Anonymised queries, minimal user data, reusable model                |
| M7  | Safety & Security               | Regular security meetings, incident reporting                        |
| M8  | OWI Licence                     | Research (default) · Commercial (OWICL) · Data Access (raw WARC)     |
| M9  | Legal & Ethical Self-Assessment | Questionnaire at download time, ethics score, signed PDF certificate |
| M10 | Ethical & Legal Advice          | Ethics workshops, legal FSTPs, OSF working groups                    |

## Three Licence Types

- **Research** — default; included in owilix
- **Commercial (OWICL)** — requires personal contact with coordinator
- **Data Access** — raw WARC; research only, personal request

## Open Web Index Licence (OWIL) – Version 1.0

Last Updated: May 8th, 2024

### Preamble

To promote an open human-centred search engine market and provide a solid basis for building front ends for offering a true choice to internet users in selection of preferable search engines, OpenWebSearch.EU develops and pilots the core of a European Open Web Index (OWI). It is the foundation for an open and extensible European Open Web Search and Analysis Infrastructure.

### Definitions

“Index Data” – The OpenWebSearch.eu Index Data consists of processed web data derived from previously crawled web data. They are processed in a way that they serve as basis for search applications and web search engines. Hence, they consist of a search index (list of terms and related web documents) and meta data of the original web documents including the plain texts and processed information. The Index Data is distributed over several data centres and organised in partitions.

# Open Webmaster Console — Transparency Tools

## Website Management

### Website Management

Explore indexed website data, verify ownership to gain management access, and submit content removal requests

SEARCH

Note: Search results reflect our database as of December 08, 2025. We're continuously updating our index to provide the most current information.

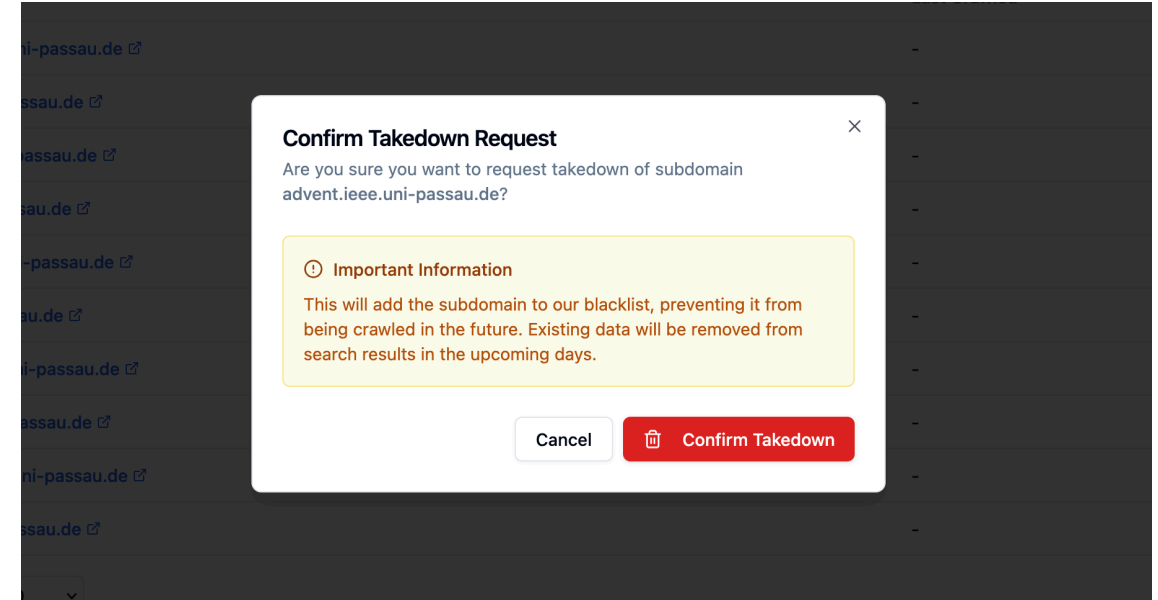
#### Recently Indexed Websites

Browse recently discovered websites in our database (225 total)

|                             |                                |
|-----------------------------|--------------------------------|
| 3zza0.xn163.com             | <a href="#">View Details →</a> |
| qroq6.szsiqi.com            | <a href="#">View Details →</a> |
| 06npb.xn163.com             | <a href="#">View Details →</a> |
| maruhide.com                | <a href="#">View Details →</a> |
| gcmu2.jimforl.com           | <a href="#">View Details →</a> |
| shop1385571536059.qiyeku.cn | <a href="#">View Details →</a> |
| zjg.haoyun56.com            | <a href="#">View Details →</a> |

- Search any domain in the OWI — crawl stats, content types, language distribution,

## Content Removal & Rights



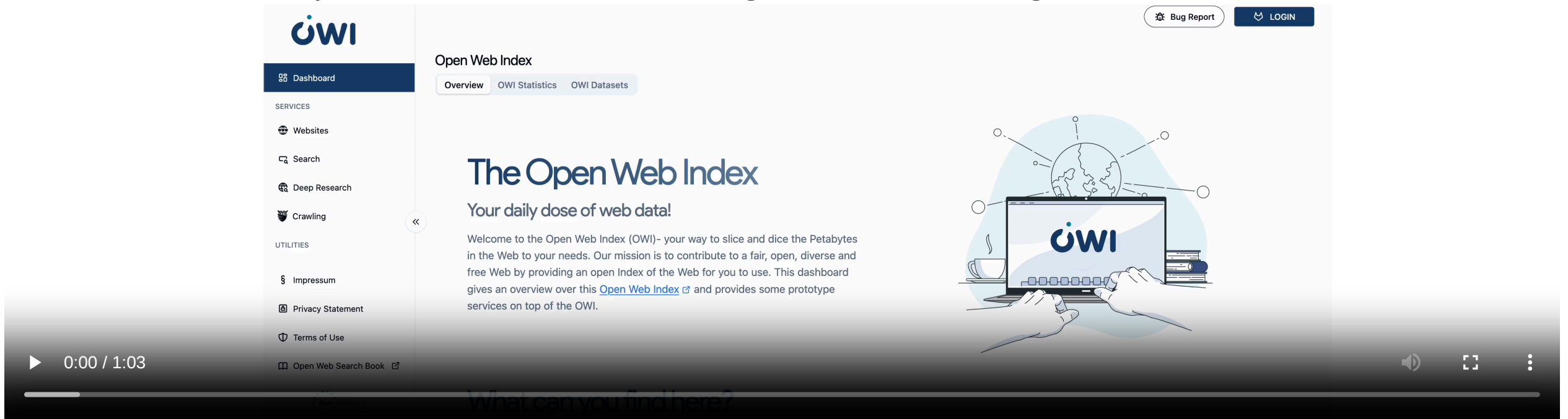
- Verified domain owners can submit **takedown requests** — subdomain or URL level
- Removed content is added to a blacklist; changelog propagated to all index users
- **Ethical self-assessment** survey required before enabling dataset downloads
- **OWLer crawler config**: express crawl constraints directly in the dashboard

subdomains

- **Domain verification:** prove ownership to unlock management capabilities

# Transparency & Webmaster Engagement

- Dashboard provides full transparency on crawling activity
- Webmasters can claim ownership of their sites
- Express usage constraints and issue takedown requests
- Gen AI exclusion protocols: robots.txt, user-agent, X-Robots-Tag



# FreeWebSearch Charter & Regulatory Reform

#FreeWebSearch Charter — launched 29 September 2025

10 Principles: transparency & explainability · competition & independent infrastructure · privacy without tracking · user/creator/advertiser control · equal access & non-discrimination · information diversity · environmental & social responsibility · integrity against disinformation · search competences & critical awareness · democratic control across borders

Four commitments: (a) serves society, people first; (b) respects democratic values; (c) transparent and open; (d) reflects internet diversity

**81 signatories** as of February 2026 — [openwebsearch.eu/charter](https://openwebsearch.eu/charter)

## 3 Regulatory Reform Proposals

1. **Copyright** — new statutory exception for open web indexes in DSM Directive
2. **Data Protection** — public-interest exemption under Art. 6(1)(e) GDPR
3. **Interoperability** — classify OWI as neutral data intermediary under DSA

Promoted via EU Data Union Strategy consultation (July 2025) and European Parliament breakfast event (December 2025)

# Societal Impact

- **Digital sovereignty** — reducing dependency on non-European search engines and AI data pipelines
- **Environmental sustainability** — reducing redundant crawling; shared infrastructure; data centre energy efficiency (CERN heat reuse)
- **Economic opportunity** — ecosystem enabling SMEs and startups; market potential study shows **ROI within 4 years**, projected **net benefit of ~4.5 B EUR over a decade**
- **Democracy & inclusiveness** — unbiased information access for informed citizens; multilingual/multicultural coverage; digital literacy

## Proposal: European Web Data Clearing House

A public infrastructure that collects, preprocesses and serves bulk web data as a certified, legally compliant data source — eliminating liability risks for European innovators at scale



# Dissemination, Communication & Exploitation

# Dissemination, Communication & Exploitation

## Dissemination

- **93 papers** published (target: 50) + 11 accepted/submitted
- **3 keynotes**, 10 invited talks, 123 talks & panels
- **#ossym** annual symposium: 2023 · 2024 · 2025 · 2026 (planned)
- WOWS workshop series at ECIR (2024 · 2025 · 2026)
- 3 data challenges · 3 hackathons · 9 webinars
- **11 master theses**, 9 bachelor theses

## Community & Stakeholders

## Communication

- LinkedIn: **882 followers** (+263%)
- Mastodon: **1,700+ followers** (+353%)
- **63 blogposts** · 12 media articles (5 countries)
- 3 podcast appearances (Deutschlandfunk, Federal Bar)
- **560 newsletter subscribers** (+224%)
- OWI Dashboard: ~1,500 unique visitors/month

## Exploitation

## Governance

- OSF mandated to coordinate project legacy
- Partners continue under MoU
- Monthly Community Meetup (post-project)
- Target: inclusion in EuroHPC AI Factory infrastructure

## Open Source

All code, pipelines, and data formats released as open source; FAIR principles on Zenodo; OWI treated as European public good

- **180+ members** on Mattermost community platform
- #FreeWebSearch Charter: **81 signatories**
- European Parliament breakfast (Dec 2025)
- Web Data Working Group at EuroHPC AI Factories
- Partnership with **EUSP** (Ecosia + Qwant)

- OSF as non-profit sustainability vehicle
- Market potential: **~4.5 B EUR net benefit** over 10 years, ROI within 4 years
- 14 partners with individual follow-on plans
- 20 open-source projects on public GitLab

# IN DETAIL

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# Scientific Dissemination

## Publications — Exceeded All Targets

| Category                     | Count      |
|------------------------------|------------|
| Journal papers               | 1          |
| Conference & workshop papers | 92         |
| <b>Total published</b>       | <b>93</b>  |
| Accepted (CHIIR 2026)        | 2          |
| Submitted                    | 9          |
| <b>Grand total</b>           | <b>104</b> |

Target was 50 papers — **186% of goal achieved**

Venues: ECIR, SIGIR, ACL, CLEF, CIKM, EMNLP, WWW, IJCAI, ICTIR etc.

## Talks & Events

- 3 keynotes · 10 invited talks · 123 talks, presentations, panels
- 11 workshops · 11 lectures · 5 lectures with tutorials · 1 seminar
- 3 data challenges · 3 hackathons · 9 webinars · 8 training events

## Theses Supervised

### Event Series

#ossym — Open Search Symposium

- 2023: inaugural
- 2024: Garching (Munich)
- 2025: CSC Espoo (Helsinki)
- **2026: Berlin** (planned)

### WOWS Workshop Series

Web-scale Open Web Search at ECIR  
— 2024, 2025, 2026 (planned). Also:  
Dagstuhl Seminar on RAG (Sept 2025)

# Communication — Organic Growth Across All Channels

**Social Media** (*organic only — no paid advertising*)

| Channel  | Followers | Growth               |
|----------|-----------|----------------------|
| LinkedIn | 882       | +263% since Feb 2024 |
| Mastodon | 1,700+    | +353% since Feb 2024 |

339 posts published since midterm report · avg. 2 posts/week

## Media Coverage

- **12 articles** in print/online media across 5 countries (Feb 2024 – Feb 2026)
- Outlets: CERN Courier, c't magazine, heise, tagesspiegel, The Register, HiPEAC magazine, European Perspectives, Bertelsmann Change Quarterly
- **3 podcast appearances:** Besser Wissen Podcast, Deutschlandfunk Kultur, (R)ECHT INTERESSANT (German Federal Bar Association)
- 63 blogposts on project website + ~18 on partner sites

## Newsletter & Website

- 560 subscribers (+224%), 5 issues since March 2024
- 59 sub-pages on openwebsearch.eu · 8 videos on Vimeo
- OWI Dashboard (launched June 2025): ~1,500 unique visitors/month · avg. visit duration **25 min 20 sec** · 316 registered users



## OWI Brand Identity

Complete brand created: logo kit, signet, brand colours, claim

**"Time to reclaim the web"**

## Print Materials

- 1,000 project brochures
- 500 OWI product flyers
- 50 Policy Paper brochures (Parliamentary Breakfast)
- 100 Economy Study copies
- 2m × 3m back-wall display

# Community Building & Stakeholder Engagement

## Community Platform (Mattermost)

- **180+ members** — public, joinable via [openwebsearch.eu/community](https://openwebsearch.eu/community)
- Channels: Tech & Application (most active), Economy, Legal
- Monthly Community Meetup continues post-project (merged Community Updates + OSF WG Tech & Application)

## Key Events

- **Public OWI Launch Event** (6 June 2025): 100+ live participants; index state, OWI licence, ethical self-assessment, use cases — recording on Vimeo
- Joint community updates with **NGI Search** (Nov 2024, June 2025)
- Index access training for FSTPs (Nov 2024): owilix + MOSAIC
- Czech-Bavarian Supercomputing Summer School 2025

## Policy Engagement

- 26 policy-level meetings · 6 member state briefings
- **European Parliament breakfast** (Dec 2025): MEPs Alexandra Geese (Greens/EFA) + Elena Sancho Murillo (S&D); Deputy DG Renate Nikolay (DG CNECT)
- Policy paper: *"Boosting digital sovereignty and competitiveness through a European Web Data Infrastructure"*
- NGI Forum 2025 · EuroHPC Summit 2025 (Krakow)



## Key Partnerships

- **EUSP** (Ecosia + Qwant joint venture) — regular consultation; joint workshop in Paris
- **OpenEuroLLM** — EU sovereign language models; interest in European-language web pages
- **JRC** — company pages for Europe Media Monitor
- **AGES** (Austria) — eMarketShield: online marketplace monitoring for nutritional supplements
- **Eurostat** & national statistical offices — web analytics use cases

## EuroHPC Web Data Working Group

Formed Summer 2025, led by LUMI and HammerHAI AI Factories — advocates for OWI as standard AI Factory infrastructure

# Exploitation Strategy & Governance

## Market Potential (Mucke Roth & Company assessment)

- **ROI within 4 years** · projected **net benefit ~4.5 B EUR over a decade**
- Business plan modelled: EUR 25k/year per customer, 30 → 356 clients over 4 years; revenue EUR 750k (2026) → EUR 7.6M (2029); break-even at month 24
- Decision: NGO+GmbH model **not pursued** due to regulatory risk and funding uncertainty; governance formalized in D6.9

## Post-Project Sustainability

- **OSF** (Open Search Foundation) mandated to coordinate project legacy
- Partners continue operating under **MoU** to bridge until AI Factory inclusion
- New organization with secured long-term funding identified as key need
- 2 new computing centres onboarded: **Exaion** (EDF Group, France) + **Universität Oldenburg** (Germany)

## Open-Source Strategy

- 20 open projects on public GitLab
- FAIR principles applied to all Zenodo records
- Top downloaded dataset: Touche23-ValueEval — **37,421 downloads**
- Top downloaded software: OWLer URLFrontier — 210 downloads

## Individual Partner Plans

Key follow-on commitments:

- **UNI PASSAU**: BMBF SOURCE (2026), startup incubation, OUR<sup>2</sup>S
- **BADW-LRZ**: OWS@LRZ, SOURCE (1.5 FTE), EuroHPC WG
- **WEBIS**: Resilipipe, CORAL, Agent CRIS, 3 PhD students
- **TUGraz**: MOSAIC frameworks, 6 master theses
- **CERN**: OWLer continuation, 2–4 PhD positions
- **OSF**: 3 FTE sustained, partnerships with Qwant & Ecosia

## Pending regulatory challenges



- WARC demand vs. EU copyright limits on raw data sharing
- Commercial licence still blocked by liability risk
- Legal certainty required for scale-up

# Conclusion

# In Summary

## What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

## Why it matters

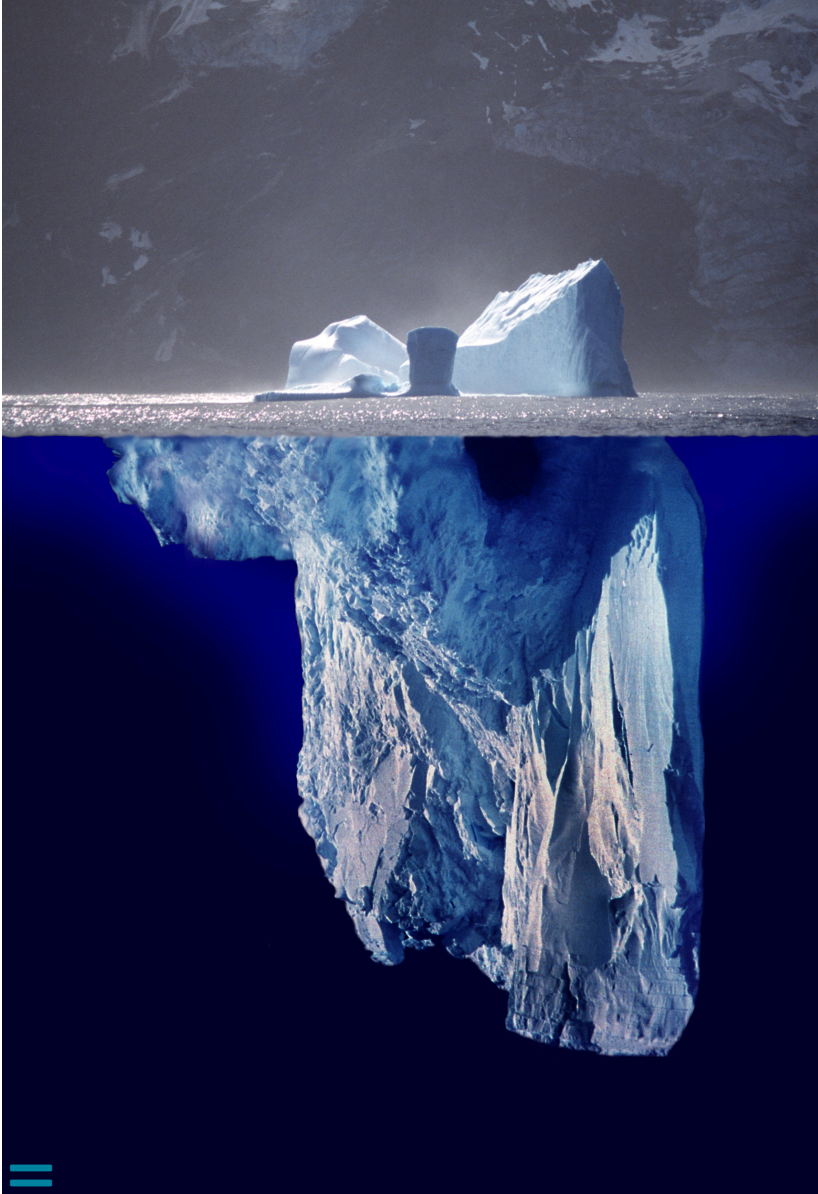
- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable

# IN DETAIL

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# Key Messages



- Web-data and web search are **critical for Europe's digital sovereignty**
- **Reducing entry barriers** for researchers, innovators, and businesses
- Opening the core element in web search: the **Open Web Index**
- Open software stack, open pipelines, open data
- Distributed, data-centric, web-scale compute infrastructure
- Web data for **AI and Retrieval Augmented Generation**
- Building an **ecosystem** around data, software, and infrastructure
- Transparency and control for legal, ethical, and societal values

## We are looking for ...

- **Researchers & tech innovators** — develop new search & retrieval paradigms and content analysis algorithms
- **Data centres** — help hosting a distributed Open Web Index
- **Industry & business partners** — discover the business models of an Open Web Index
- **Policy makers** — help shaping the governance of an open search ecosystem

## Contact

- Email: [join@openwebsearch.org](mailto:join@openwebsearch.org)
- Community: [community@openwebsearch.eu](mailto:community@openwebsearch.eu)
- Website: [openwebsearch.eu](https://openwebsearch.eu)
- Dashboard: [openwebindex.eu](https://openwebindex.eu)
- Mattermost: [openwebsearch.eu/community](https://openwebsearch.eu/community)

**Thanks. Questions? Comments? Feedback?**



# Before we come to the Questions

# We are looking for ...

## Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

## Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

# Get in touch

|                 |                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
|                 |                                                                                                                 |
| Website         | <a href="https://openwebsearch.eu">openwebsearch.eu</a>                                                         |
| Dashboard       | <a href="https://openwebindex.eu">openwebindex.eu</a>                                                           |
| Code Repository | <a href="https://opencode.it4i.eu/openwebsearcheu-public/">https://opencode.it4i.eu/openwebsearcheu-public/</a> |
| Community       | <a href="https://openwebsearch.eu/community">openwebsearch.eu/community</a>                                     |
| Join            | <a href="mailto:join@openwebsearch.org">join@openwebsearch.org</a>                                              |
| General         | <a href="mailto:community@openwebsearch.eu">community@openwebsearch.eu</a>                                      |

Thank you — Questions, Comments, Feedback very welcome!